

AERO62520 Coursework 6 Final Design Review

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Declaration of originality

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1 Introduction

1.1 Problem Statement Summery

Global industries—from logistics and agriculture to disaster response—are confronting acute labor shortages and mounting pressure to improve operational efficiency, creating an urgent need to automate repetitive, hazardous, and resource-intensive tasks. Future automation must go beyond simple point-to-point motion and instead combine autonomous navigation, real-time object recognition, and precise manipulation to operate reliably in unstructured, unpredictable environments. This project responds to that need by targeting tasks that reduce human exposure to risk while increasing throughput and adaptability in dynamic settings.

The objective is to design and develop an autonomous rover capable of independently exploring unknown terrain, identifying target blocks via computer vision, retrieving them with an integrated robotic arm, and delivering loads to designated collection points. Achieving this requires tight coordination among sensing, planning, and actuation subsystems, and the system is intended to simulate real-world industrial collaboration scenarios where robustness and repeatability are essential. Key subsystems include SLAM for localization and mapping, inverse kinematics for accurate arm control, and a mechanically stable chassis sized for expected payloads.

1.2 Addressed Feedback Summery

The following section (Table 1) summarizes the feedback received during the Preliminary Design Review (PDR) and outlines the modifications and refinements implemented in response to enhance the overall design and meet specified requirements.

Table 1. Addressed Feedback Summary

Subject Area	Feedback	Response
System Block Diagram 5/5; Component description and justification 5/5	System block diagram is good. Justification good and linked to requirements.	Accepted. The team maintained the clear hierarchical structure praised in the PDR. The diagram has been updated in the FDR (see Figure 1) to reflect the transition to USB-C PD power architecture and wired communication while maintaining high clarity. The methodology of directly linking selection criteria to functional requirements has been retained. This standard has been applied to the newly selected Anker batteries (see end of Table 3 to ensure they meet mission requirements.
Manufacturability and structural strength 4/5	Manufacturability considered, some structural strength calculation would be beneficial.	Agreed, although only certain components merit structural analysis e.g. components with significant loading or where stability is critical for function. Dynamic analysis could provide additional insight although all structural concerns can be easily mitigated in this instance through the addition of material at minimal financial and sustainability cost due to manufacturing method. Yield Strength and Axial Stiffness calculations added for the Lidar Mount as the most structurally critical element.
CAD 4/5	CAD well done with extensive breakdowns. Extra dimensions of parts would be more beneficial.	Agreed, extra dimensions could help communicate the mechanical design and relative sizes. However, critical dimensions were specified and typically if a full breakdown was required, technical drawings would be requested. Additional dimensions added.

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Table 1. Addressed Feedback Summary (Continued)

Subject Area	Feedback	Response
Electrical diagram 5/5; Power 5/5	Electrical diagram is good. The choice and description of the power problem is well made. The proposed solution is too heavy (marks retained).	Agreed. Although the previous solution met sustainability criteria, it barely met dimensionality constraints and pushed total system mass to nearly 8 kg, degrading performance. Our new solution uses 2 × Anker Prime 26K with a 1.2 m 20 V/5 A USB C PD→5.5 mm cable for the NUC and a 1.2 m 12 V/5 A USB C PD→5.5 mm cable for the manipulator. Battery mass is reduced from ≈4.17 kg to ≈1.20 kg, a reduction of ≈2.97 kg (≈71.3%) relative to the earlier battery baseline; relative to the 2.82 kg allowance the reduction is ≈1.62 kg (≈57.4%). Total payload with the new batteries (components 1.879 kg + sled 0.30 kg + batteries 1.20 kg = ≈3.38 kg) is well below the rover's nominal 5.00 kg capacity. Despite supplying the NUC at 100 W PD rather than a 120 W peak, expected mission profile compute loads remain within the delivered power and runtime for the 35 minute profile should be unchanged. Supporting electrical diagram is also modified accordingly.
Software block diagram 3/5	Software diagram a bit too detailed, I think you should remove topics name and stick to packages.	Agreed. Topic names were removed from the diagram to simplify the presentation and focus on the package-level structure. The node architecture has also been updated to better reflect the current system design and improve clarity of the software organisation.

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Table 1. Addressed Feedback Summary (Continued)

Subject Area	Feedback	Response
Mission planning through software 2/5	Mission planning considered briefly in bringup node.	Based on the previously designed logic, the mission flow has been clarified and divided into separate operational states. Robot behaviour is controlled through a state machine implemented in the bringup node, where transitions are triggered by topic callbacks and action completion callbacks. This structure allows clearer coordination between navigation, perception, and manipulation modules. This will be explained more clearly in words apart from the block diagram.
Git documentation 0/5	Link to github not included	Agreed. A link to the project repository has now been included for documentation and code

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Table 1. Addressed Feedback Summary (Continued)

Subject Area	Feedback	Response
Tests 3/10	Tests implicitly defined, you should define exactly how are you testing for each requirement.	There was some confusion as to whether the verification methods should remain solution independent (given this for the requirements that they are verifying). Furthermore, it is my understanding that explicit testing details would be captured in a separate document (a Test Plan) which was never requested as a deliverable in the PDR Brief. Furthermore, Table D1 of the NASA handbook only specifies the testing method as Inspection/Demonstration/Test/Analysis. Furthermore, there was no mention of “tests” anywhere within the brief, so it was assumed that this was not expected output (at least explicitly in this context). Clarification was sought via email with Dr Ladosz. Following this, the assumptions that very brief test plans were required for each requirement was made and added as additional detail to the requirements verification matrix. It is assumed that this is the extent of the expectation.
Test achievability 3/5	Tests do not have specific objectives, but they are generally achievable in context of robotics	As Above.
Success criteria 3/5	Success criteria only defined as part of requirements not as part of the test	As Above.

1.3 Feedback Modifications Summary

Following the feedback from the Preliminary Design Review (PDR), Table 2 outlines the subsequent design refinements and modifications. Central to these changes was the implementation of the new Anker Prime twin battery arrangement and a complete mechanical redesign, which prioritizes sustainability and manufacturability (detailed in Section 3 Mechanical Design). These updates were cascaded through the rest of the system, resulting in key improvements in power-to-weight optimization, structural integrity through stress analysis, and more detailed software architecture and verification processes to guarantee requirement compliance.

Table 2. Feedback Modifications Summary

Subject Area	Feedback	Modifications
System Block Diagram 5/5; Component description and justification 5/5	System block diagram is good. Justification good and linked to requirements.	Modified the relevant diagram and table based on the battery selection for the Anker Prime 26K.
Manufacturability and structural strength 4/5	Manufacturability considered, some structural strength calculation would be beneficial.	Yield Strength and Axial Stiffness and Beam Displacement calculations added for the Lidar Mount in the mechanical section.
CAD 4/5	CAD well done with extensive breakdowns. Extra dimensions of parts would be more beneficial.	Rudimentary dimensioned technical drawings have been added for key dimensioned components to the mechanical section of the report.

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Table 2. Feedback Modifications Summary (Continued)

Subject Area	Feedback	Modifications
Electrical Diagram 5/5; Power 5/5	Electrical diagram is good. The choice and description of the power problem is well made. The proposed solution is too heavy.	Replaced heavy pack with $2 \times$ Anker Prime 26K; battery mass reduced from ≈ 4.17 kg $\rightarrow \approx 1.20$ kg (-2.97 kg, -71.3%), bringing total payload to ≈ 3.38 kg (< 5.00 kg). Supporting electrical diagram modified accordingly.
Software block diagram 3/5	Software diagram a bit too detailed, I think you should remove topics name and stick to packages.	Remove topic names, and modified node structure based on current plan
Mission planning through software 2/5	Mission planning considered briefly in bringup node.	Based on previous designed logic, dividing the process into separate states and controlling the robot behaviour using state machine which triggered by topic callbacks and action done callbacks.
Git documentation 0/5	Link to GitHub not included.	Will include this time: https://github.com/longchuanYoRHa/Group12_team_space.git
Tests 3/10	Tests implicitly defined, you should define exactly how are you testing for each requirement.	Tests briefly specified in the Verification Methods column of the Requirements Verification Matrix. More granular assessment criteria have been added to the grade column providing a benchmark for the pass/marginal/fail criteria. It is assumed that this is the extent of the expectation.

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Table 2. Feedback Modifications Summary (Continued)

Subject Area	Feedback	Modifications
Test achievability 3/5	Tests do not have specific objectives, but they are generally achievable in context of robotics.	Specific objectives and success metrics assigned to each test case.
Success criteria 3/5	Success criteria only defined as part of requirements not as part of the test.	Tests briefly specified in the Verification Methods of the Requirements Verification Matrix.
Communication	Not mentioned in PDR feedback.	To enhance stability, the communication method between the NUC to Rover and manipulator has been modified from Wi-Fi to an Ethernet cable connection.

2 System Design

2.1 System Block-Diagram

The hardware connections of the system are shown in Figure 1.

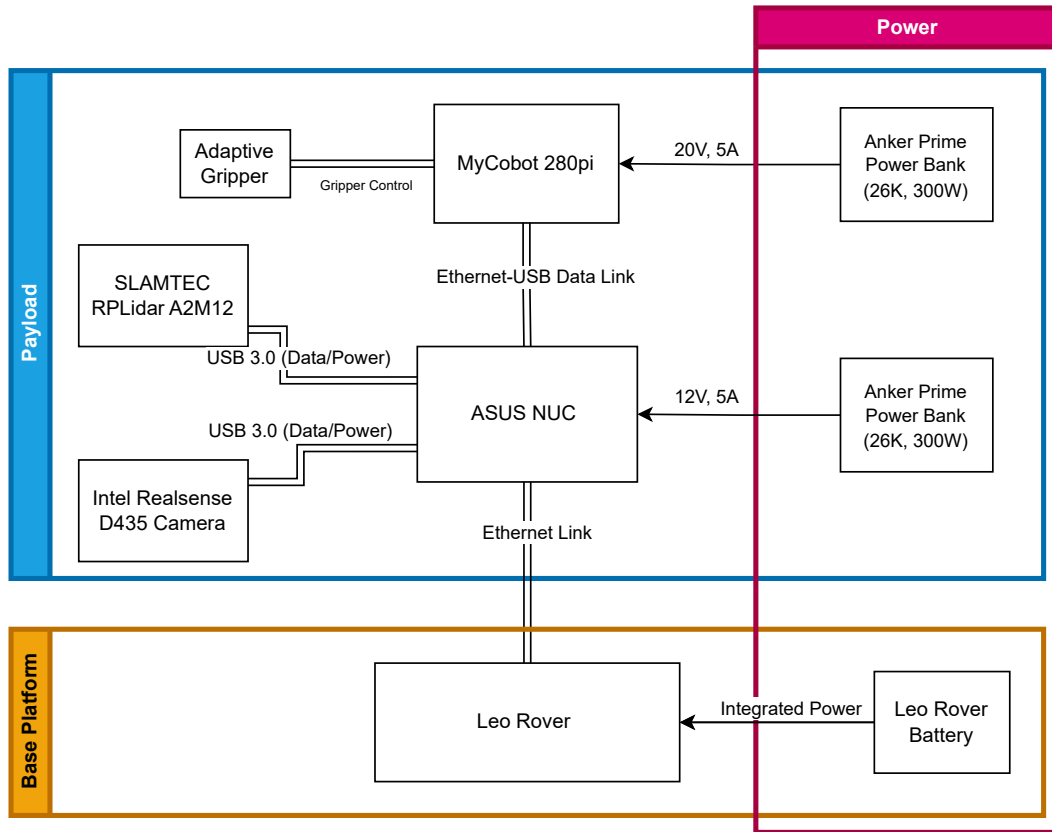


Fig. 1. System Architecture and Power Distribution Block Diagram

2.2 Component Selection Matrix

Table 3 below presents detailed descriptions and justification for each core component.

Table 3. System Component Specifications and Selection Rationale

Component / Subsystem	Key Specifications	System Function	Justification & Design Rationale
Leo Rover 1.8 (Mobility Platform)	• Payload: 5 kg (Nominal) • Drive: 4WD, Independent Suspension • Core: Raspberry Pi 4B	Provides mobility capabilities to transport the payload payload across the arena and navigate unstructured terrain.	Payload Matching: The 5 kg capacity sufficiently supports the NUC, manipulator, and auxiliary battery (Total ≈ 4.6 kg). Terrain Adaptability: Independent suspension ensures stability when traversing obstacles or cable guards in the lab environment.
ASUS NUC 12 Pro (Computational Core)	• CPU: i7-1260P (12 Cores) • RAM: 16 GB • I/O: USB 3.2 Gen 2	Executes high-load algorithms including SLAM navigation, YOLOv8 object detection, and Movelt trajectory planning.	Compute Demand: Only x86 architecture with high clock speeds can satisfy the real-time inference requirements (> 30 FPS) for YOLO, which exceeds Raspberry Pi capabilities. Bandwidth: High-speed USB ports ensure simultaneous data ingestion from LiDAR and RGB-D cameras without bottlenecks.
Intel RealSense D435 (Vision Perception)	• Tech: Active Stereo IR • Range: 0.3 m – 3 m • Feature: Global Shutter	Provides RGB streams for object classification and aligned depth clouds for 3D spatial coordinate extraction.	Motion Robustness: The Global Shutter sensor minimizes motion blur during robot movement, ensuring reliable color block detection. Positioning: The depth data allows direct deprojection of 2D pixels to 3D (x,y,z) coordinates essential for inverse kinematics (FR-09).

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Table 3. System Component Specifications and Selection Rationale (Continued)

Component / Subsystem	Key Specifications	System Function	Justification & Design Rationale
SLAMTEC RPLIDAR A2M12 (LIDAR Navigation)	• Range: 12 m • Sample Rate: 16k/sec • Resolution: 0.9°	Delivers 360° environmental scanning data for 2D mapping (SLAM) and obstacle avoidance.	Coverage: The 12 m range fully covers the CW3 arena dimensions. Stability: High sampling rate prevents map drift during rapid rotations, meeting the autonomous navigation requirements.
myCobot 280 Pi (Manipulator)	• DOF: 6-Axis • Payload: 250 g • Repeatability: ± 0.5 mm	Executes end-effector manipulation tasks: approaching, grasping, transporting, and depositing target blocks.	Dexterity: 6-DOF design allows approach from arbitrary angles, adapting to different pickup heights. Capacity: 250 g payload exceeds the maximum block weight of 150 g (PR-04). Lightweight: Low self-mass (850 g) minimizes the impact on the rover's center of mass.
Adaptive Gripper (End Effector)	• Stroke: 20 mm – 75 mm • Interface: LEGO Compatible • Control: Digital I/O	Physically grasps objects of varying sizes securely.	Size Compatibility: The stroke range perfectly covers the 20 mm standard blocks and potential larger obstacles. Simplicity: Simple I/O control reduces software complexity compared to dexterous hands, increasing reliability.

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Table 3. System Component Specifications and Selection Rationale (Continued)

Component / Subsystem	Key Specifications	System Function	Justification & Design Rationale
2 X Anker Prime 26K (Additional Power)	• Capacity: 26,250 mAh • Output: 20V (DC), 5A, 100W and 12V (DC), 5A, 60W • Weight: ≈ 1.2 kg	Provides independent power to the NUC and Manipulator, isolating the load from the rover base.	Reliability: Separation of power ensures that high computational loads do not drain the mobility battery. Integration: Direct DC output allows the use of proprietary adapters for NUC and Arm, simplifying voltage regulation safety compliance.

3 Mechanical Design

3.1 Design Overview

Full Assembly Diagram

Figure 2 illustrates the overall layout, the relative positions of key components, and the spatial relationship between the load pallet and the joint's working space.

Following the provision of the twin Anker Prime 300w power solution a complete mechanical re-design was necessary (V2) since the previous version (V1) was built around the assumption that our specified power solution (Anker C300) would be viable. This report goes some way to compare the different versions as a method of highlighting the design improvements.

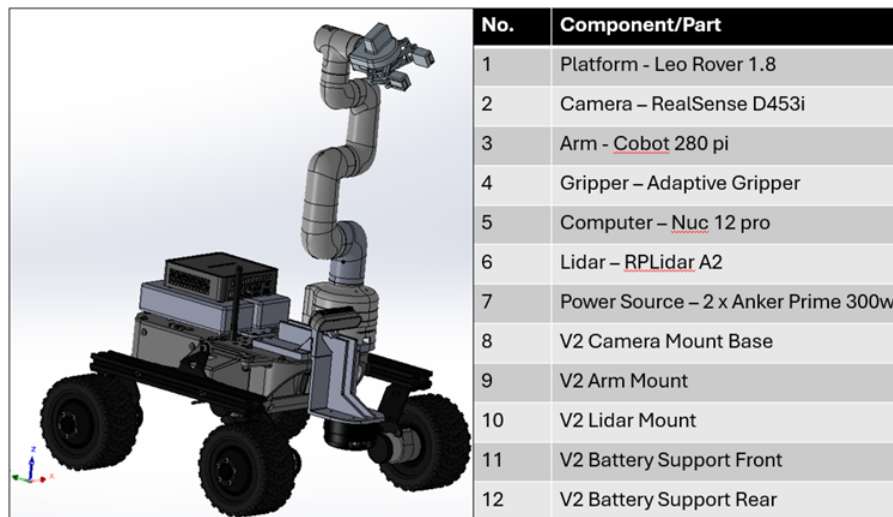


Fig. 2. Overall Assembly (Isometric). Shows the main hardware modules and component list.

Figure 3 clarifies the mounting interfaces, installation heights, and relative positioning of each component, facilitating subsequent design verification and assembly process documentation for the load pallet, camera field of view, and LiDAR field of view.

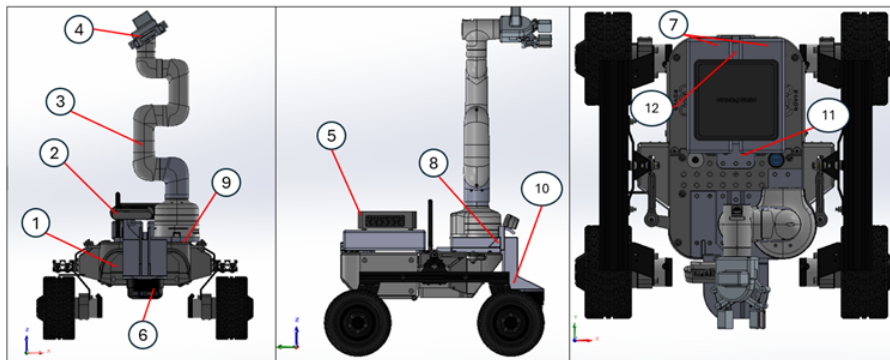


Fig. 3. Three-view drawings with component annotations (Front / Side / Rear). Part numbers correspond to the parts list for assembly and manufacturing instructions.

General Design Table

Table 4 summarizes key decisions in this robotic mechanical design: material and process selection, modular assembly strategies, manufacturability and safety considerations, and mass estimation for the payload pallet. This information supports subsequent detailed assembly drawings and manufacturing documentation.

Table 4. General Features and Design Rationale

General Feature	Description and Rationale
Material Usage	• V2 uses exclusively 3D printed parts in an effort to streamline manufacturing and improve sustainability. V1 required Lazer cutting which is a high energy process and no longer necessary. • All parts can be made from PLA which is fully recyclable and biodegradable (industrial process). • There was a significant attempt to reduce the volume of material used (V1 – V2) to reduce operational energy consumption and manufacturing cost and sustainability.
Modular Design	• V2 Components also mechanically interlock and locate onto the main body and each other. This was done to save time in assembly and provide increased structural strength. • All electrical components are fastened with M4 × 20 screws consolidating fixing type. • The design allows a level of flexibility of mounting locations, with the arm and the camera positions being interchangeable. This extends the lidar with minor modifications. • Allows redesign/repair of single subsystem/mount if required.

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Table 4 – continued from previous page

General Feature	Description and Rationale
Component Location	• Due to modular design component placement is not 100% fixed and alterations possible with minor modifications. • A general attempt to balance the robot and maintain the centre of mass has been made. Lidar: • Lidar position driven by goal of capturing the 200 mm cubed Coloured Object Plinths and greater robustness to low level obstacles than a top mounted position. • V2 has a 160° front and 40° rear field of view to ensure that the navigation subsystem can collect the required data. • This is a reduction on V1 (200°) but enables the robot to position itself closer to the target. It also reduces the stresses on the mounting arrangement due to reduced bending moments. Camera: • Camera positioned for forward approach to Coloured object plinths. • Central position for optimal viewing of coloured blocks mindful of robotic task space. Arm: • Frontal position to provide adequate grasping range, and off-centre to reduce camera occlusion during grasping and transportation. Power Source: • Mounted centrally on the rear on the robot to maintain stability and act as a counterweight to the other components. NUC: • Mounted in an easily accessible location with clearance for cables and cable management. • Unobstructed airflow for thermal regulation.
Future Proofing	• If required, additional mounted components can be stacked on top of the existing mounts (space allowing) and secured simply with longer fasteners due to the interlocking design. • The Lidar Mount design provides an interface for additional 3D printed Block storage if required (see component breakdown for details). • Additional 3D printed parts may be added for cable management.

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Table 4 – continued from previous page

General Feature	Description and Rationale
General Robustness	• More material than structurally necessary has been used to mitigate potential damage caused by drops, transport, and improper storage. • Minimum part thickness of 5 mm.

3.2 Part Breakdown

V2 Camera Mount Assembly - Part 8

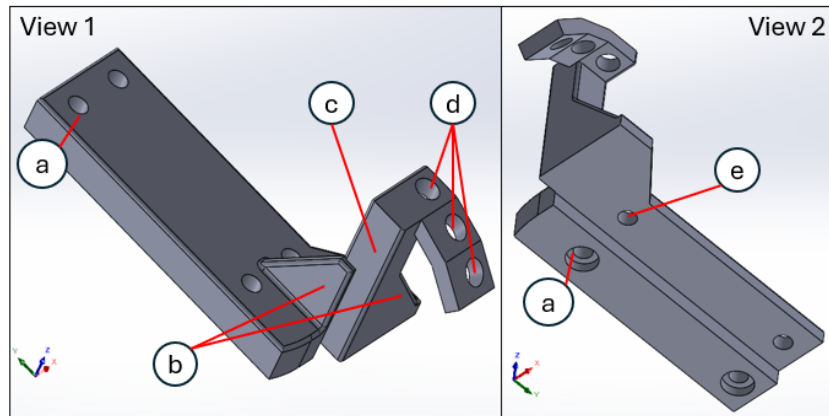


Fig. 4. Camera Mount Perspective Detail Views (Part 8).

The camera mounting assembly, as shown in Figure 4 and Table 5, is manufactured using 3D printing technology with PLA material and is suitable for printing in any orientation.

Table 5. Key Features and Rationale of the Camera Mount Assembly

Feature	Description and Rationale
a	M5 through holes with interference fit (0.1mm) recesses for locating to rover mounting points.
b	5mm thick reinforcement ribs to improve structural support.
c	Structural post.
d	M6 through holes for camera mounting positioned at 0° , -20° , -40° .
e	M5 through hole for fastening to V2 Lidar Mount.

V2 Arm Mount – Part 9

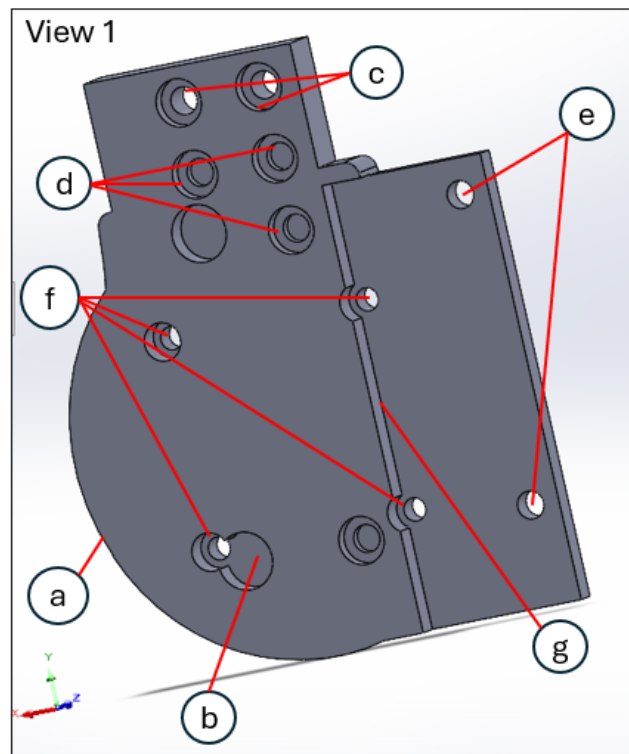


Fig. 5. Arm Mount Perspective Detail Views and Assembly (Part 9).

The arm mount, as shown in Figure 5 and Table 6, is manufactured using 3D printing with PLA material and is fabricated in an off-plane orientation.

Table 6. Key Features and Rationale of the Arm Mount Design

Feature	Description and Rationale
a	External profile fully supports arm base.
b	Clearence recess for Rover Fixings.
c	M5 through holes with interference fit (0.1mm) recesses for locating to rover mounting points.
d	Clearence recess for Rover Fixings.
e	M5 through hole for fastening to V2 Lidar Mount.
f	Access holes to enable fastening of MyCobot280 pi to Arm Mount.
g	Interface for locating to V2 Lidar Mount.

V2 Lidar Mount – Part 10

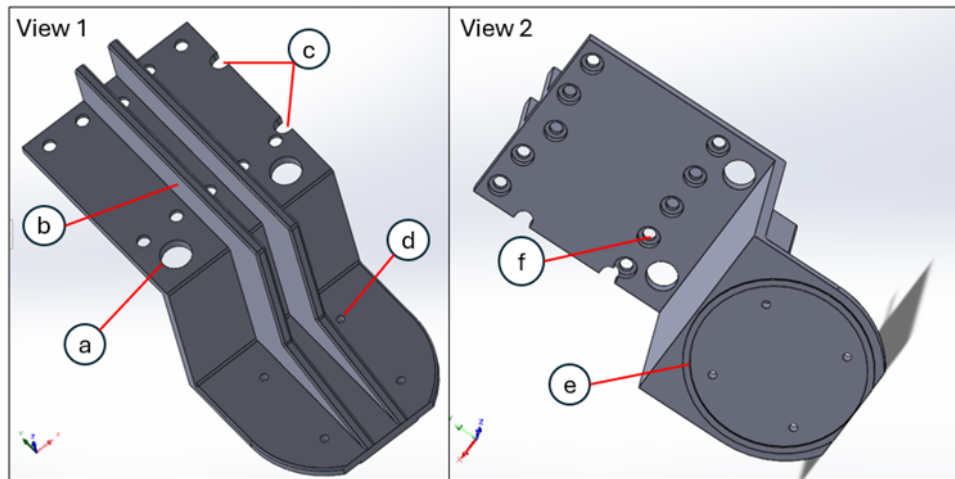


Fig. 6. Lidar Mount and Rear Support Perspective Detail Views and Assembly (Part 10).

The lidar mount, as shown in Figure 6 and Table 7, is manufactured using 3D printing with PLA material and is fabricated in an off-plane orientation.

Table 7. Key Features and Rationale of the Lidar Mount Design

Feature	Description and Rationale
a	Clearance recess for Rover Fixings.
b	Twin structural webs to increase strength of part and reduce dynamic effects when the rover is in motion. Functions as a locating interface for other parts. Also acts as a mounting point for potential coloured object transportation bin.
c	Clearance holes for M4 Arm mounting screws.
d	M3 through hole for fastening Lidar to mount.
e	Locating ring to assist in assembly of Lidar-Mount.
f	M5 through holes with interference fit (0.1mm) recesses for locating to rover mounting points.

V2 Battery Support Rear – Part 11

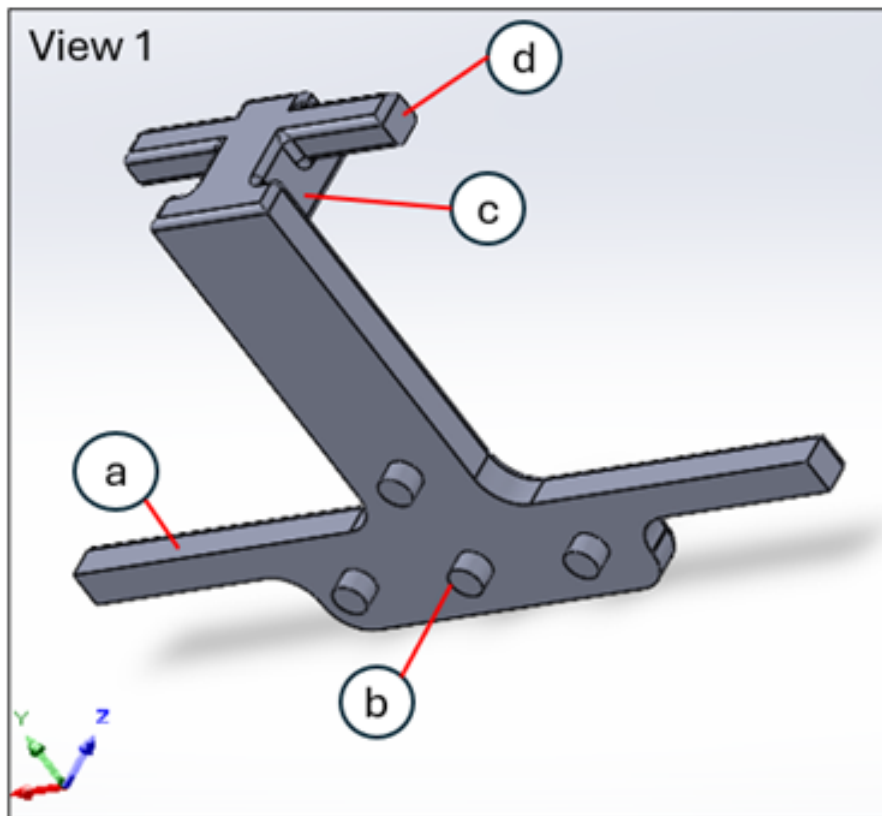


Fig. 7. Battery Support Rear Perspective Detail Views and Assembly (Part 11).

The rear support, as shown in Figure 7 and Table 8, is manufactured using 3D printing with PLA material and is suitable for printing in any orientation.

Table 8. Key Features and Rationale of the Rear Support

Feature	Description and Rationale
a	Battery rest support arm. Large chamfers to reduce stress concentration points.
b	Locating pins that interface with the rover chassis.
c	Spacer and structural post.
d	Rear arms to locate batteries at rear of rover.

V2 Battery Support Front – Part 12

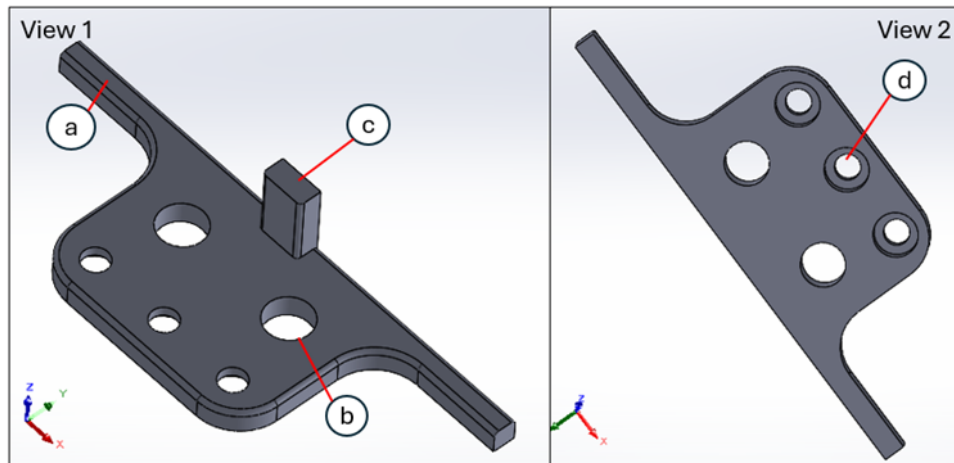


Fig. 8. Battery Support Front Perspective Detail Views and Assembly (Part 12).

The front support, as shown in Figure 8 and Table 9, is manufactured using 3D printing with PLA material and is suitable for printing in any orientation.

Table 9. Key Features and Rationale of the Battery Support Front Design

Feature	Description and Rationale
a	Battery rest support arm. Large chamfers to reduce stress concentration points.
b	Rover fixing clearance holes.
c	Spacer to locate batteries.
d	M5 through holes with interference fit (0.1mm) recesses for locating to rover mounting points.

Additional Part 1 – Camera Arm Test Rig Plate

Additional Test Rig plate for Rover independent testing of the Arm and Camera. It interfaces with the mounting plates to create a testing platform with known special transforms. As shown in Figure 9 and Table 10, is manufactured using 3D printing with PLA material and is suitable for printing in any orientation.

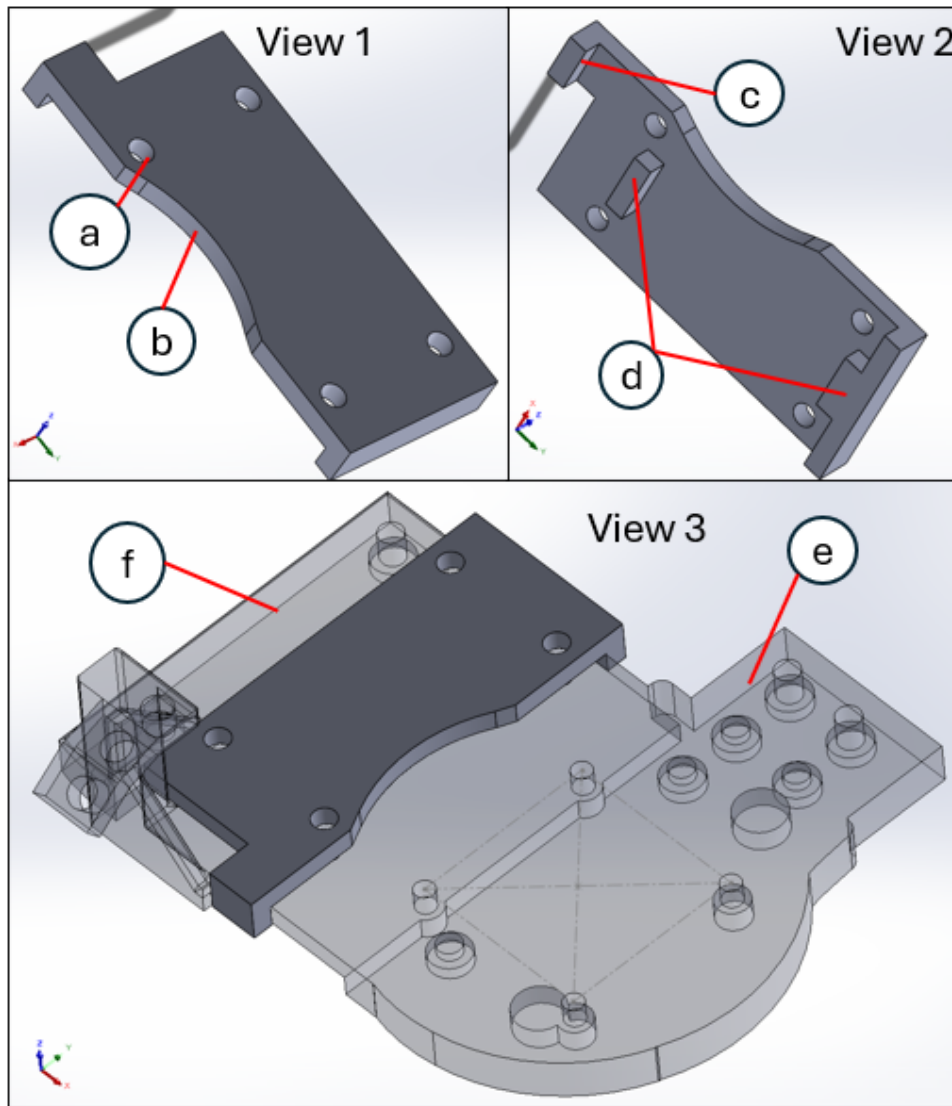


Fig. 9. Camera Arm Test Rig Plate (Additional Part 1).

Table 10. Key Features and Rationale of the Main Platform Design

Feature	Description and Rationale
a	M5 through holes for securing the 2 other component mounts.
b	Cutout for Arm clearance.
c	Locating plates.
d	Spacers.
e	Arm Mount.
f	Camera Mount.

Fixings

Table 11 details the complete bill of materials for the mechanical fasteners used in the assembly, including their specific applications. M4 fixings have been used where not constrained by existing components.

Table 11. Summary of Required Fixings and Fasteners (BOM)

Qty.	Fixing Type	Application / Notes
10	M5 × 20 Screw	Arm mount, Camera mount, Lidar mount, and Battery support front to Rover body.
4	M4 × 10 Screw	Arm to mount.
4	M3 × 10 Screw	Lidar to mount.
1	M6 × 10 Screw	Camera to mount.

Total Mass of Sled

The total mass of the V2 payload sled is 304 g which is a 76% decrease in mass. The infill of 80% was assumed as a middle ground due to the likelihood of a significantly lower infill selected for printing and the prevalence of high surface area to volume parts with higher edge density. A detailed mass breakdown of individual components is presented in Table 12.

Table 12. Mass and Volume Analysis for V2 Structural Components

Metric	Value	Units
Volume of Parts	304.610	cm ³
Density of PLA	1.24	g/cm ³
Assumed Density (80% Infill)	1.00	g/cm ³
Total Mass of V2 Parts	304.61	g
Total Mass of V1 Parts (Reference)	1275	g

Calculations

As a result of feedback from the PDR, structural stress calculations were added. Since the Lidar mount is the only structural element this is the only one considered. In all instances the worst case was considered.

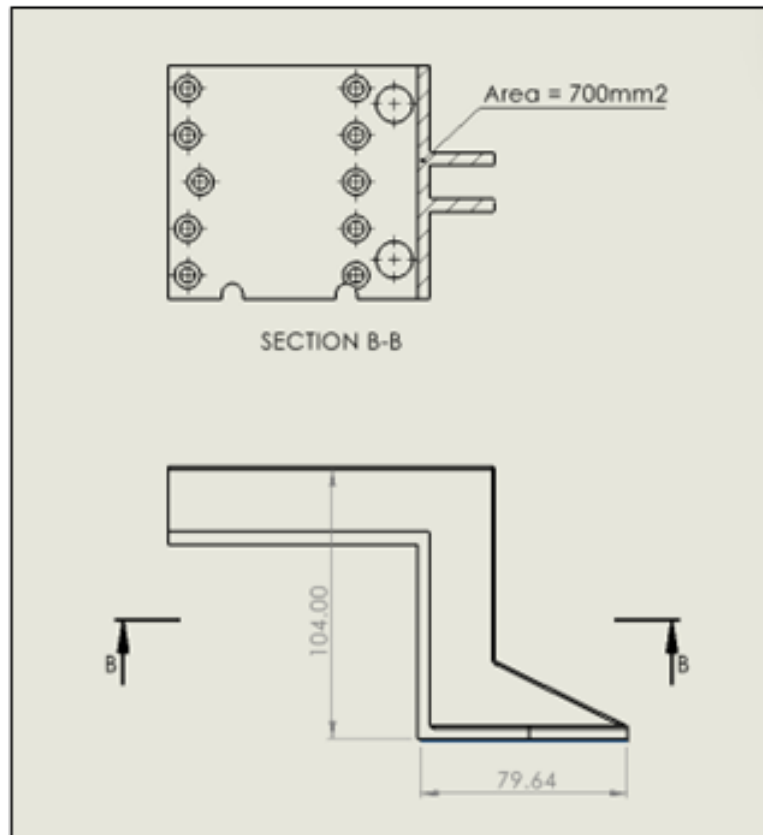


Fig. 10. Cross Section B-B showing the calculation area (Area = 700 mm²).

Table 13. Lidar Mount – Yield Strength (z axis)

Metric	Value	Units
Mass of Lidar	190	g
Mass of Lidar Mount	163	g
Mass Total (M)	353	g
Yield Strength PLA (0.2%el)	30	MPa
Cross Sectional Area (A)	700	mm ²
Gravity (g)	9.8	m/s ²
Force (F) = $M \times g$	3.43	N
Stress (σ) = F/A	4,900	Pa (N/m ²)
Factor of Safety (FoS) = Y_s/σ	6.12	

Table 14. Lidar Mount – Static Axial Deflection (z axis)

Metric	Value	Units
Youngs Modulus (E)	2.5	GPa
Cross Sectional Area (A)	700	mm ²
Length (L)	104	mm
Axial Stiffness (k) = AE/L	1.68×10^8	N/m
Static Axial Deflection = F/k	1.9×10^{-8}	m
	1.9×10^{-5}	mm

Table 15. Lidar Mount - Deflection (Euler Bernouli – End Loaded Cantaliver)

Metric	Value	Units
Youngs Modulus (E)	2.5	GPa
Force (F) (from previous)	3.43	N
Length (L)	79.64	mm
Second Moment of Area (I_{xx}) (T section)	50119	mm ⁴
Deflection = $(FL^3)/(3EI)$	4.61×10^{-4}	m
	0.0461	mm

Given these values, even with significantly greater forces from dynamic effects it is very unlikely to significantly displace the lidar.

Further Analysis

Structural analysis was deemed unnecessary given the scope of the project, relatively static loading, and material thickness.

FEA might be informative for locating stress concentration points and guiding hole positioning, given that parts of this nature are often susceptible to brittle fracture.

The orientation of layers in the 3d prints is important in maintaining the strength of the parts and ideally should be printed off plane.

Drawings

Due to feedback form the PRD rudimentary technical drawings of the Arm, Camera, and Lidar were produced.

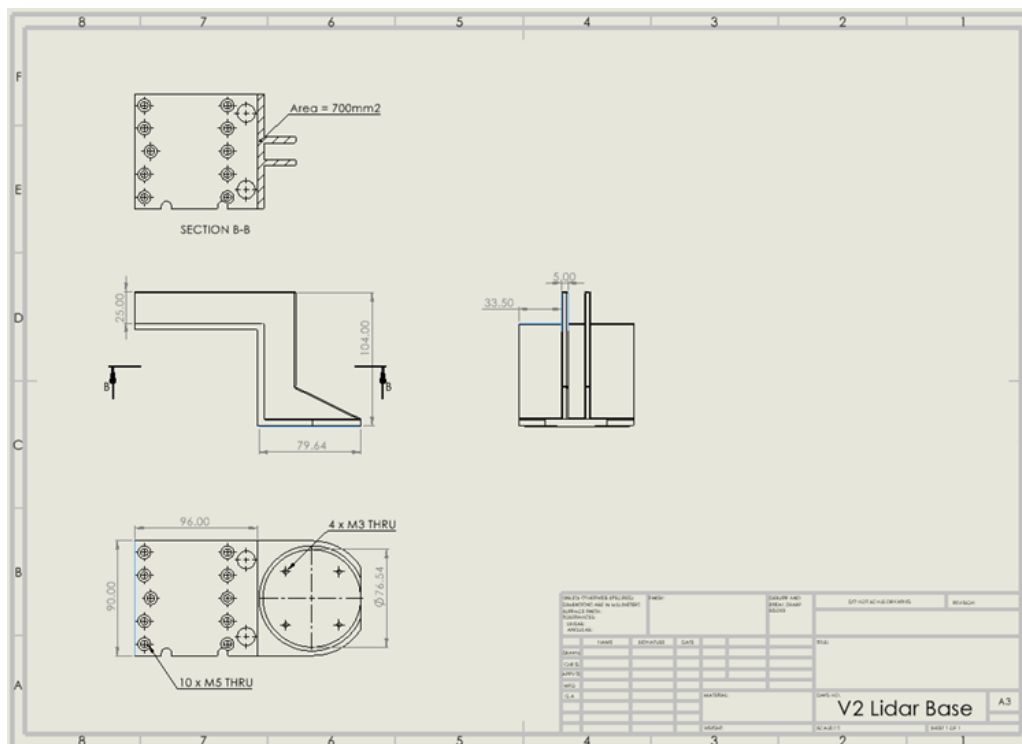


Fig. 11. Technical Drawing of the V2 Lidar Base.

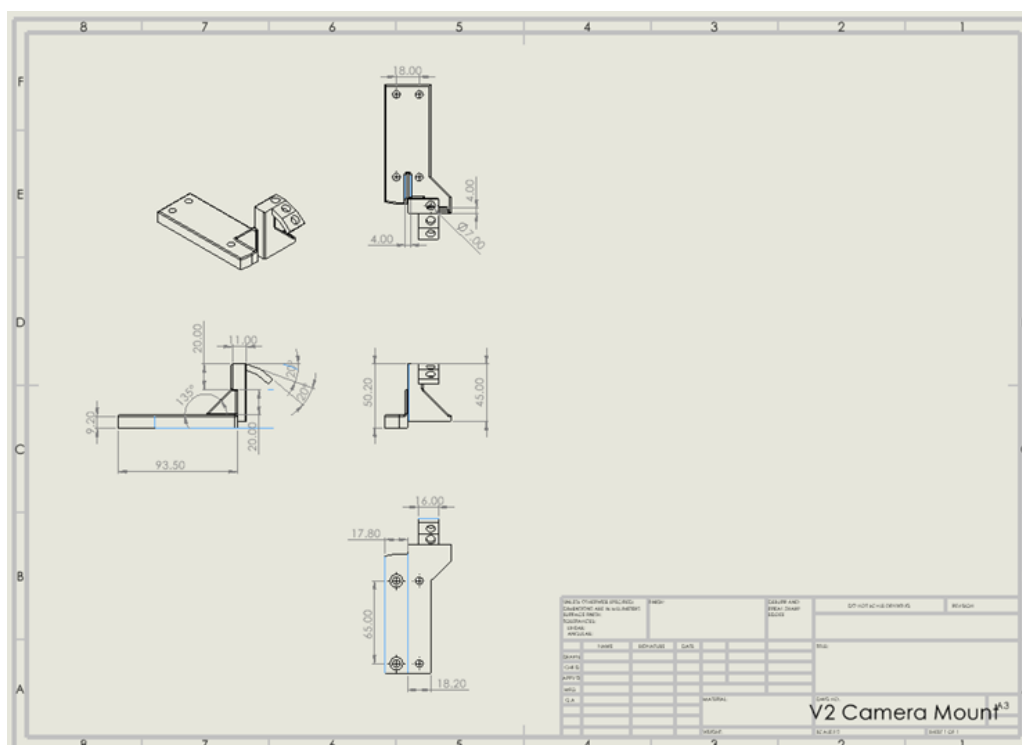


Fig. 12. Technical Drawing of the V2 Camera Mount.

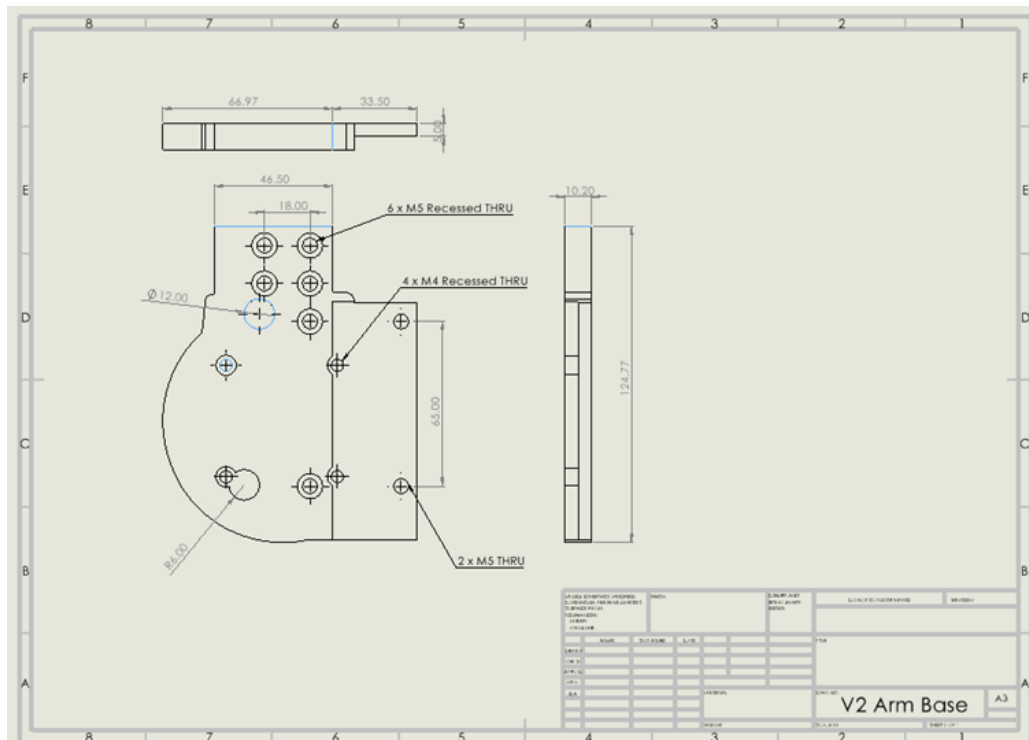


Fig. 13. Technical Drawing of the V2 Arm Base.

4 Electrical Design

4.1 Mission Scope and Duration

The onboard power system shall be sized to support a complete operational cycle of the rover: autonomous navigation and arena scanning using onboard cameras, obstacle detection and wall-margin avoidance using the LIDAR, manipulation of a target block (including approach, grasp, transport back to the base, and placement into the designated bin), and all associated sensing, computation, telemetry, and safety functions.

The power supply must reliably sustain at least one full cycle of this mission loop and provide continuous untethered operation for a minimum of 20 minutes, as specified by stakeholder requirements.

4.2 Theoretical Power Draw on Component Power Draw Specifications and Mission Scope

Component power consumption specifications and theoretical power consumption for the task scope are shown in Table 16.

Table 16. Component Power Consumption Estimation

Component	DC Volt (V)		Current (A)		Wattage (W)		Dur. (Hr)	Energy (Wh)	
	Min	Max	Min	Max	Min	Max		Min	Max
Asus NUC (Compute)	12	20	Unsp.	10	Unsp.	120	0.33	-	39.96
myCobot 280pi	12	12	5	5	60	60	0.33	19.98	19.98
myCobot 280pi Adaptive Gripper	Unsp.	5	Unsp.	Unsp.	-	-	0.33	-	-
Intel RealSense D435i	4.75	5.25	Unsp.	0.700	Unsp.	3.66	0.33	-	1.21
RPLiDAR A2M12	5	5	0.45	0.60	2.25	3.00	0.33	0.74	0.99
Leo Rover 1.8	11.1	Unsp.	Unsp.	8	Unsp.	Unsp.	0.33	-	-

Note: Unsp. = Unspecified. Wattage calculated as Current $\times$ Voltage. Data derived from component specifications.

4.3 Practical Design Considerations and Calculations

Practical design considerations and calculations are shown in Table 17.

Table 17. System Power Budget and Engineering Rationale

Item(s)	Parameter / Metric	Value / Calculation	Engineering Design Choice Rationale
Mission Scope	Min. Adjusted Duration	35 min (0.583 h)	The duration includes a conservative 75% operational headroom applied to the estimated runtime. This crucial safety measure is implemented to ensure practical battery sufficiency and mitigate the risk of mission failure by absorbing contingencies such as: task retries, power draw inconsistencies stemming from component age or specification variances, and real-world battery degradation.
Rover Base	Power Requirement	None	The base is capable of operating independently on its integrated battery for a minimum of 1 hour, thus no external power source is required for its operation.
Power Architecture	Topology	Either: (A) Single large power station with two AC outputs feeding existing adapters (NUC adapter and manipulator adapter) OR (B) Two PD power banks: Battery A → NUC/peripherals (PD 20 V / 5 A, 100 W); Battery B → Manipulator (PD 12 V / 5 A, 60 W)	Design keeps options open for final procurement and test results, and enables direct comparison of weight, dimensionality, and approval timelines.
NUC (ASUS)	Max Power Draw (Rated)	120 W	Retained as the NUC's rated maximum for assessing performance under heavy compute loads (SLAM, vision inference).

Continued on next page...

Table 17. System Power Budget and Engineering Rationale (Continued)

Item(s)	Parameter / Metric	Value / Calculation	Engineering Design Choice Rationale
	Delivered Power (Load)	100 W (20 V × 5 A)	PD cable and negotiation limit delivered power to 100 W. For expected workloads, performance is expected to remain largely unchanged.
	Assumed PD + Efficiency	95%	Conservative allowance for PD negotiation, cable losses, and internal DC-DC conversion losses in the NUC and peripherals.
	Operating Current (at 20 V)	5.0 A	Matches PD cable rating and maintains safe current limits.
	Power from Battery	105.26 W (100 W/0.95)	Battery capacity is sized to cover conversion losses and provide margin for PD negotiation variability.
36	Depth Camera	Power Draw (Est.) 3.5 W (5 V × 0.7 A)	Maximum USB current constraint; supplied from NUC USB Type-C port on the 20 V PD rail.
	LiDAR	Power Draw (Allocated) 2.5 W	Allocated 0.5 A at 5 V from the NUC USB Type-A port; fits within the NUC USB current budget.
	LiDAR + Camera	Peripherals Source Direct NUC USB	Both draw from NUC's regulated 5 V rails. Each port supports up to 1.5 A, ensuring sufficient current with margin.
	NUC + Peripherals	Total Energy Demand 58.3 Wh (100 W × 0.583 h)	Calculated on a continuous basis (not duty-cycle) as these must run uninterrupted for SLAM and visual processing.
		Total Energy from Battery 61.4 Wh (105.26 W × 0.583 h)	Battery must supply this to guarantee mission completion including all conversion losses.

Continued on next page...

Table 17. System Power Budget and Engineering Rationale (Continued)

Item(s)	Parameter / Metric	Value / Calculation	Engineering Design Choice Rationale
Manipulator	Delivered Power (PD Rail)	60 W (12 V $\times$ 5 A)	PD cable rating matches servo and gripper requirements; accounts for possibility of exceeding expected duty cycles.
	Assumed PD + Efficiency	95%	Conservative allowance for PD negotiation and DC-DC conversion losses inside the manipulator adapter.
	Power from Battery (Rail)	63.16 W (60 W/0.95)	Accounts for conversion losses so battery capacity is sized correctly for the rail.
Gripper	Power Source	5 V from Manipulator	Does not require a separate power source. Actuation occurs only when servos are idle to ensure full wattage is allocated.
Manipulator + Gripper	Total Energy Demand	34.98 Wh (60 W $\times$ 0.583 h)	Continuous-run assumption provides a conservative budget for the 35-minute profile.
	Total Energy from Battery	36.8 Wh (63.16 W $\times$ 0.583 h)	Total energy required from the dedicated manipulator power bank to ensure reliability.

4.4 Power Budget Analysis

The detailed power budget breakdown is presented in Table 18.

Table 18. System Power Budget Breakdown

Source / Adapter	Power to Load (W)	Assumed Adapter Efficiency (%)	Power from Battery (W)	Component	Load Power (W) ($V \times I$)	Duration (h)	Energy at Load (Wh)	Energy from Battery (Wh)
Battery A / Source Port A → NUC (PD 20V/5A)	100	95	105.26	ASUS NUC	94 (20 V x 4.7 A)	0.583	58.30	61.37
				Depth Camera	3.5 (5 V x 0.7 A)			
				LiDAR	2.5 (5 V x 0.5 A)			
Battery B / Source Port B → Manipulator (PD 12V/5A)	60	95	63.16	myCobot 280pi Manipulator + Gripper	60 (12 V x 5 A)	0.583	34.98	36.82
Totals	—	—	—	—	160	0.583	93.28	98.18

Notes: Adapter efficiency of 95% is used for PD cable + internal conversion losses (PD negotiation and DC-DC conversion inside the device). Energy from battery = (Power from Battery) × Duration. Load energy = (Load Power) × Duration.

The Leo Rover base station battery has a capacity of 5800 mAh, a nominal voltage of 11.1 V DC, and a maximum output current of 8 A. Based on these specifications, the estimated minimum continuous operating time is 1 hour.

4.5 Power Source Constraints

Supply Constraints

- **NUC (PD) supply requirement:** Must deliver **100 W PD** (20 V × 5 A) to the ASUS NUC rail to support the NUC, Intel RealSense depth camera, and SLAMTEC LiDAR. Per-rail energy requirement for a 35-minute mission: ≈ 61.4 Wh (includes PD negotiation and conversion losses).
- **Manipulator (PD) supply requirement:** Must deliver **60 W PD** (12 V × 5 A) to the myCobot manipulator rail for servo motor and gripper operation. Per-rail energy requirement for a 35-minute mission: ≈ 36.8 Wh (includes PD negotiation and conversion losses).
- **Combined PD energy requirement:** System energy required from batteries for a 35-minute mission is ≈ 98.2 Wh (NUC ≈ 61.4 Wh; Manipulator ≈ 36.8 Wh). Each PD bank must provide the per-rail usable Wh specified.

- **Safety and compliance:** PD banks and PD → barrel cables must provide stable output and reliable PD negotiation at the required voltages/currents; include appropriate overcurrent protection and verify connector and cable ratings under continuous load .

Weight Constraints

- **Payload limit:** The rover's nominal payload capacity is constrained to **5.00 kg**.
- **Known component mass:** Combined mass of the NUC, depth camera, LiDAR, manipulator, gripper, and wiring totals **1.879 kg**.
- **Payload sled:** Additional payload sled mass $\approx$ **0.3 kg or 300 g**.
- **Maximum allowable battery mass:** $5.00 \text{ kg} - 1.879 \text{ kg} - 0.3 \text{ kg} = \mathbf{2.82 \text{ kg}}$. Chosen PD banks must respect this battery mass upper bound.

Dimensionality Constraints

- **Battery mounting area:** Allocated footprint for batteries within the rover base is **18.34 cm (width) × 15.3 cm (length)**.
- **Mounting rules:** PD banks must fit within this footprint or be mounted with controlled overhang that does not shift the rover's centre of mass rearward; the battery must not introduce a rearward bias that would render the rover back-heavy.

4.6 Pugh Matrix for Power Source Selection

Criteria Definition:

- **Energy Requirements Compliance** – Implementation meets all specified energy sustainability thresholds.
- **Weight Requirements Compliance** – Implementation adheres to maximum allowable weight constraints.
- **Dimensionality Requirements Compliance** – Implementation conforms to the prescribed spatial constraints and defined margins.
- **Stakeholder Requirements Compliance** – The solution meets stakeholder requirements (preference for non-tethered solutions).
- **Quality of Implementation** – The degree to which the solution achieves intended functionality reliably.
- **Cost of Implementation** – Total expenditure required for design, prototyping, and deployment.
- **Implementation Complexity** – The relative difficulty of integrating the solution into existing systems.

- **Time for Research and Implementation** – The duration required to complete feasibility studies, prototyping, and deployment.
- **Time for Approval** – The expected period for institutional validation.
- **Safety Compliance** – Conformance with all applicable safety standards and risk-mitigation protocols.

Following evaluation of the available options in Table 19, the Anker Prime 26K satisfies all defined criteria while presenting the fewest compromises. Accordingly, it has been selected as the default power source.

Table 19. Pugh Matrix for Power Source Selection (Updated)

Criteria	Wt.	Current		Alt. 1		Alt. 2		Alt. 3		Alt. 4	
		2x Anker Prime 26k		Anker Solix C300X		Jackery Exp. 240v2		Direct Tethering		Custom Source	
		B	W	B	W	B	W	B	W	B	W
Energy Requirements	5	5	25	5	25	4	20	5	25	3	15
Weight Requirements	5	5	25	3	15	5	25	5	25	5	25
Dimensionality Req.	4	5	20	4	16	3	12	5	20	4	16
Stakeholder Req.	5	5	25	5	25	5	25	1	5	5	25
Quality of Impl.	4	5	20	5	20	5	20	5	20	3	12
Cost of Impl.	2	3	6	4	8	4	8	5	10	5	10
Impl. Complexity	3	5	15	5	15	4	12	3	9	2	6
Time for Res. & Impl.	3	5	15	5	15	5	15	5	15	2	6
Time for Approval	4	5	20	4	16	4	16	5	20	2	8
Safety Compliance	5	5	25	5	25	5	25	1	5	2	10
Total Weighted Score			196		180		178		154		133

Note: Wt.: Weight (1–5); B: Base Score; W: Weighted Score. Current: 2 × Anker Prime 26k with USB-C PD cables. Alt. 1: Former best (Solix C300X).

4.7 Verdict

The proposed solution (2 × Anker Prime 26K + 1.2 m 20 V/5 A PD→5.5 mm cable for the NUC + 1.2 m 12 V/5 A PD→5.5 mm cable for the manipulator) meets the supply constraints by providing the required 100 W PD to the NUC rail and 60 W PD to the manipulator rail and satisfies the per-rail energy needs for a 35-minute mission (NUC ≈ 61.4 Wh; Manipulator ≈ 36.8 Wh; combined ≈ 98.2 Wh); it meets the weight constraint because the two Anker units combined (1.20 kg) are well below the maximum allowable battery mass (2.82 kg); it meets the dimensionality constraint since two smaller banks can be arranged within the 18.34 cm × 15.3 cm footprint without inducing a rearward COM bias; and it meets safety/compliance requirements when used with the specified 20 V/5 A and 12 V/5 A cables plus inline overcurrent protection and bench verification of PD negotiation. thus, this one is selected.

4.8 Power Connection Diagram

The specific power connections between the Payload Sied and the Leo Rover Base are shown in Figures 14 and 15

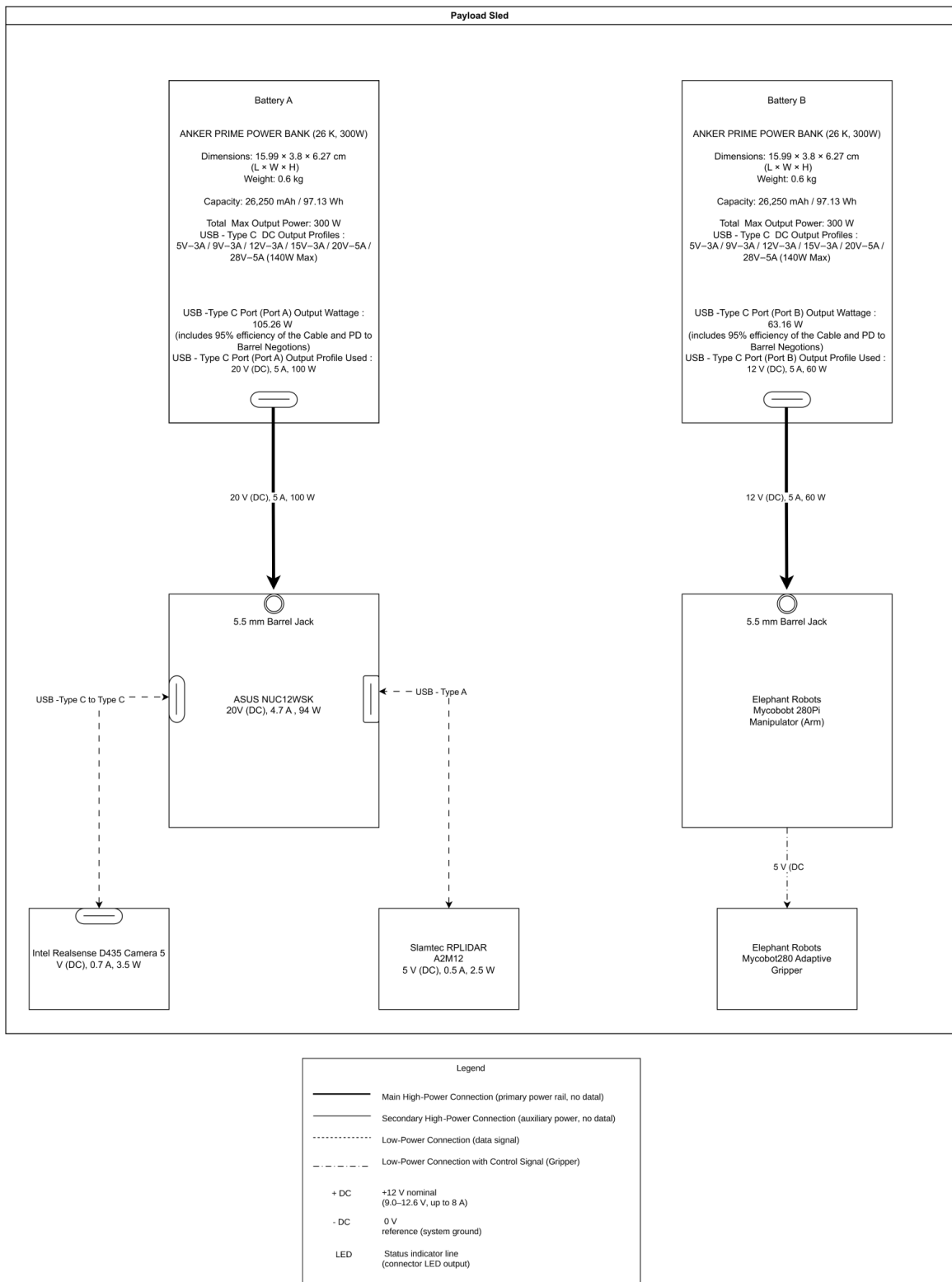


Fig. 14. Power Connection Diagram

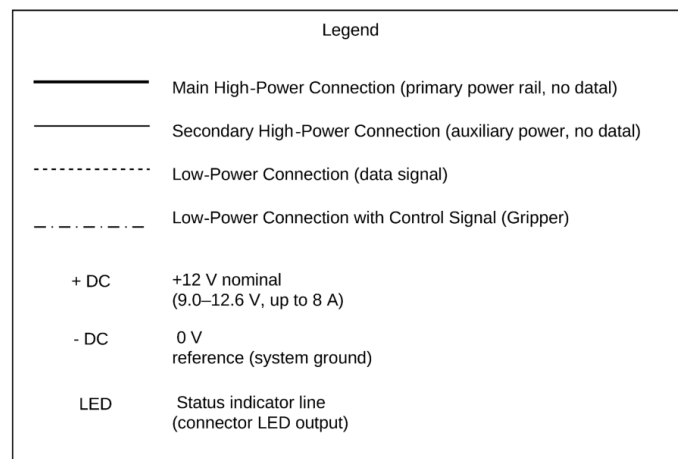
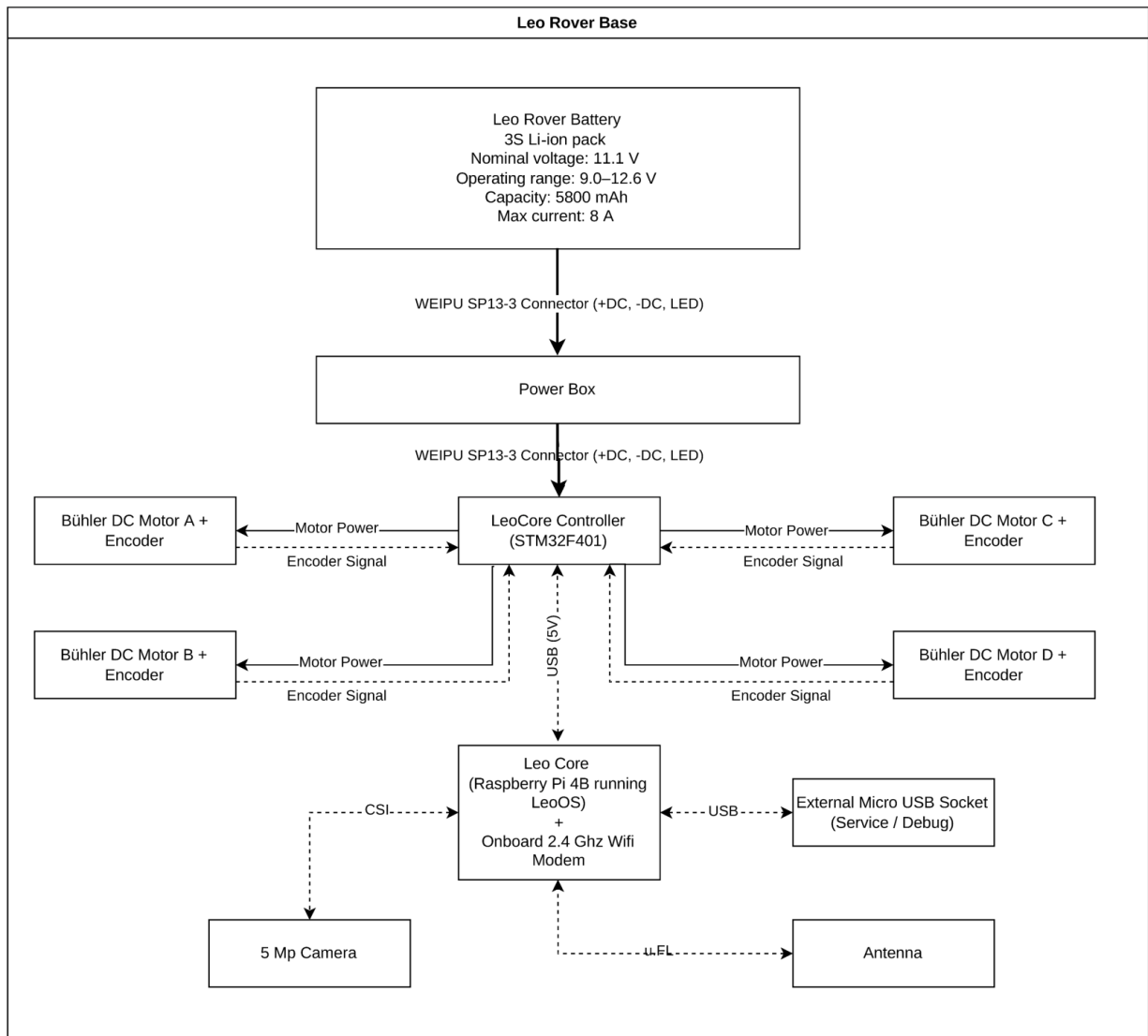


Fig. 15. Power Connection Diagram

5 Software Design

5.1 System Software Design Block Diagram

The software architecture is designed as a modular, decentralized framework implemented within the ROS 2 environment, with the Intel NUC serving as the primary computational host. As illustrated in Figure 16, the system integrates specialized hardware drivers for the RealSense D4531 and RPLidar A2M12 with high-level functional blocks, including *slam_toolbox* for mapping and the Nav2 stack for autonomous navigation.

The central controller node orchestrates the mission sequence by interfacing with peripheral subsystems—such as the Leo Rover firmware and the mycobot manipulator—through a standardized hierarchy of ROS 2 topics, services, and actions. This architecture ensures robust data flow and facilitates automated node deployment upon system startup, allowing for seamless coordination between perception, exploration, and actuation module.

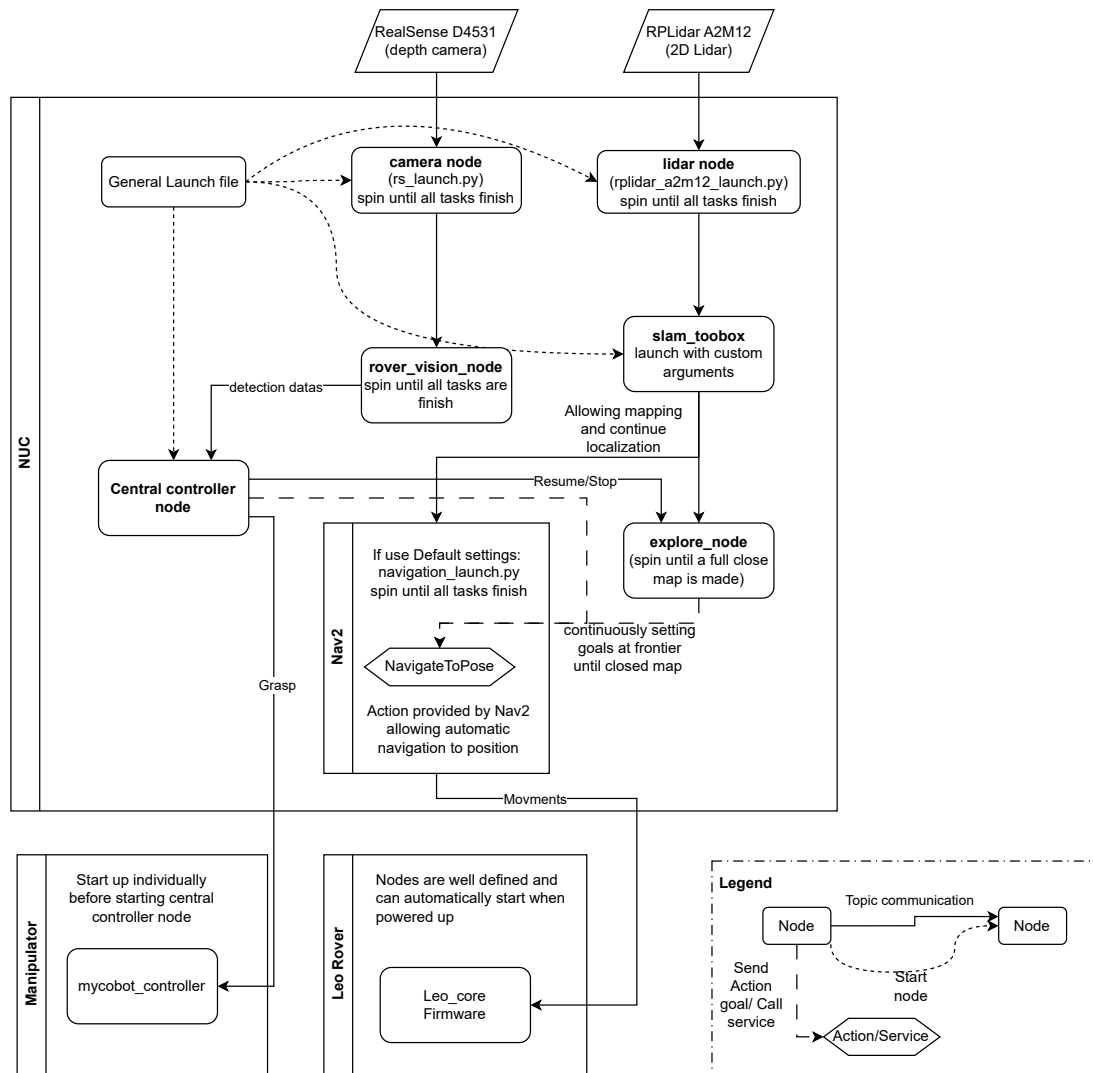


Fig. 16. System Software Design Block Diagram.

Note: Well known predefined nodes will be only showing the connection relations, detail algorithm logics will not be show in the diagram, the logic for central controller node and general process will be shown in another graph.

5.2 Central Control State Machine

The central orchestration of the software system is managed by a core control node residing on the NUC. This node implements a Finite State Machine (FSM) to coordinate high-level autonomous behaviors by processing asynchronous data and transitioning the robot's state based on real-time feedback.

As illustrated in Figure 17, the system transitions through discrete operational phases—including *EXPLORE*, *GRASP*, and *PLACE_IN_BIN*—triggered by specific conditions such as object detection callbacks, arm status updates, and navigation completion signals. The FSM architecture ensures mission robustness by incorporating error-handling logic, such as interest point blacklisting upon timeouts, and maintaining separate logic flows for loaded and unloaded operational modes to optimize task efficiency.

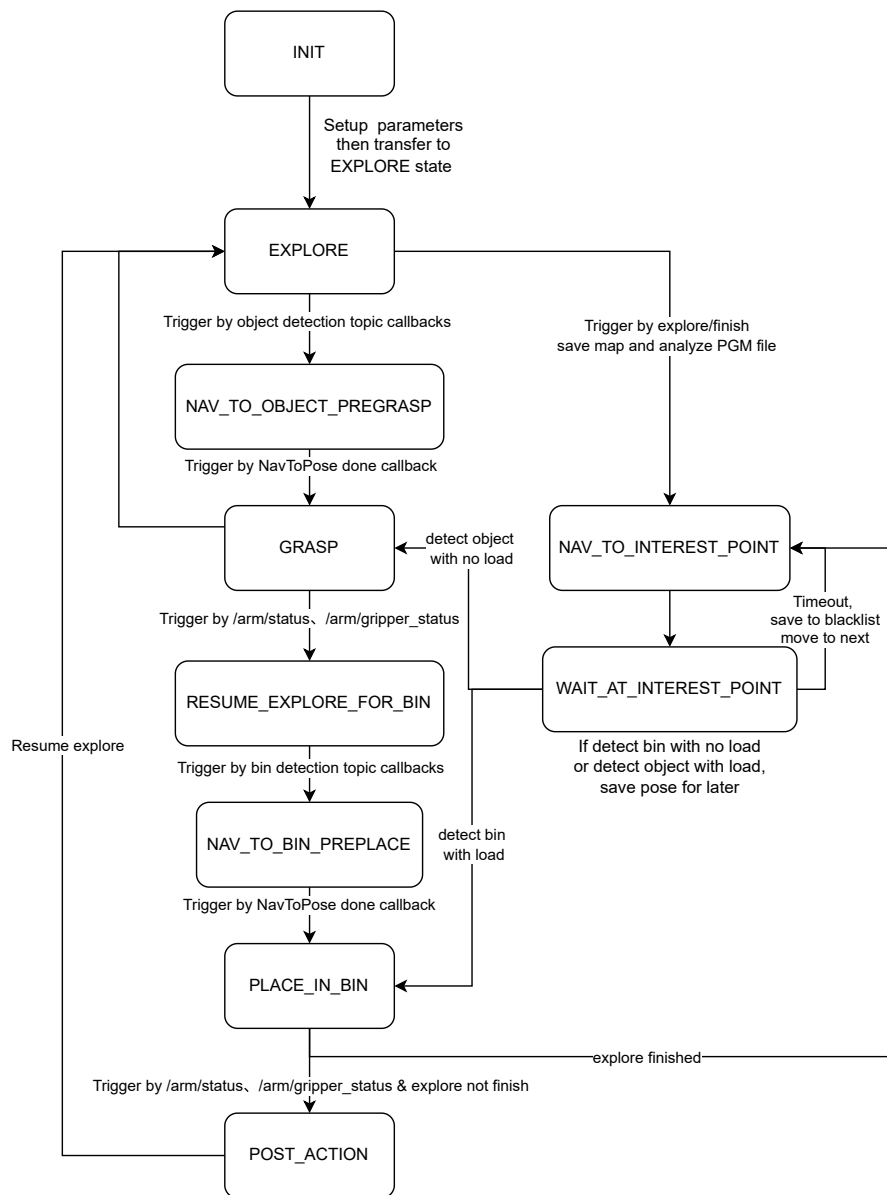


Fig. 17. Finite State Machine (FSM) diagram illustrating mission orchestration and autonomous state transitions.

5.3 Model Evaluation and Vision Logic

5.3.1 Quantitative Benchmarking

To determine the optimal object detection architecture for the NUC-based robotic platform, a comprehensive benchmarking analysis was conducted. We evaluated YOLOv8n, YOLO11n, and RT-DETR-L across four dimensions: Speed, Accuracy, Resource Efficiency, and Feature Set.

Table 20. Model Evaluation Matrix with Weighted Scores

Metric	Weight	YOLOv8n	YOLO11n	RT-DETR-L
Speed (FPS)	35%	9.0 (58 FPS)	9.0 (58 FPS)	3.0 (13 FPS)
Accuracy (Conf)	30%	7.5 (0.89)	9.0 (0.94)	7.0 (0.87)
Resource Usage	25%	9.0 (Very Low)	9.0 (Very Low)	2.0 (Very High)
Features	10%	8.0	10.0 (Enhanced)	8.0
Total Score	100%	8.45	9.10	4.45

As illustrated in Table 20, YOLO11n emerged as the superior choice with a total weighted score of 9.10. It delivers an exceptional balance of high-speed inference (58 FPS) and robust detection accuracy (0.94 confidence), significantly outperforming the Transformer-based RT-DETR-L in the resource-constrained NUC environment.

5.3.2 Technical Implementation and Hardware Acceleration

The high performance of the vision system is a direct result of specialized software optimization within the `RoverVisionNode`:

- **OpenVINO & GPU Leveraging:** The YOLO11n model is deployed using the OpenVINO inference engine. By explicitly setting the device to `intel:gpu`, the system offloads computation to the Intel integrated GPU, achieving 58 FPS while maintaining low CPU overhead.
- **Optimized Inference Pipeline:** To ensure real-time responsiveness for the manipulator, the image size is standardized to 640×480 , and a high confidence cutoff of 0.85 is enforced to minimize false positives in dynamic environments.

5.3.3 Spatial Reasoning and Size Validation Logic

The core innovation in the vision implementation is the integration of a Spatial Size Validator, which extends the standard YOLO11n features through three key steps:

1. **Sensor Alignment:** The node utilizes `rs.align` to synchronize the depth stream with the color frames from the RealSense D435i camera. This ensures that the center of every detected bounding box (c_x, c_y) corresponds to an accurate distance measurement Z .
2. **3D Deprojection:** Using the camera's intrinsic parameters, the system deprojects 2D pixel coordinates into 3D camera space. By calculating the physical distance between the left and right edges of the bounding box at a given depth, the system derives the real-world width of the object.
3. **Intelligent Categorization:** This spatial logic allows the robot to distinguish between objects of the same color but different functions. Objects with a width near 2cm are categorized as CUBES (Pick targets), while those near 20cm are identified as BINS (Place targets).

5.3.4 Dynamic Topic Routing and Safety Gatekeeping

Validated coordinates (X, Y, Z) are processed through a strategic Topic Routing mechanism:

- **Namespace Segmentation:** Based on the color prefix and validated size, coordinates are published to specific ROS2 topics (e.g., `/target_pick/red`). This decouples the vision system from the task state machine, enhancing modularity.
- **Operational Safety:** Before publication, the node enforces a `MAX_Z_LIMIT` of 0.50m. Any target beyond the physical reach of the MyCobot 280pi is automatically rejected, serving as a software "gatekeeper" to prevent mechanical overextension.

5.4 RQT Graph and System Architecture Alignment

To validate the practical implementation of the software architecture and verify the communication reliability between nodes, an RQT graph with active topic statistics was generated during a live operational cycle (Figure 18). This runtime visualization provides a clear, verifiable link to the theoretical System Block Diagram (Figure 1) and the Software Design Block Diagram (Figure 16).

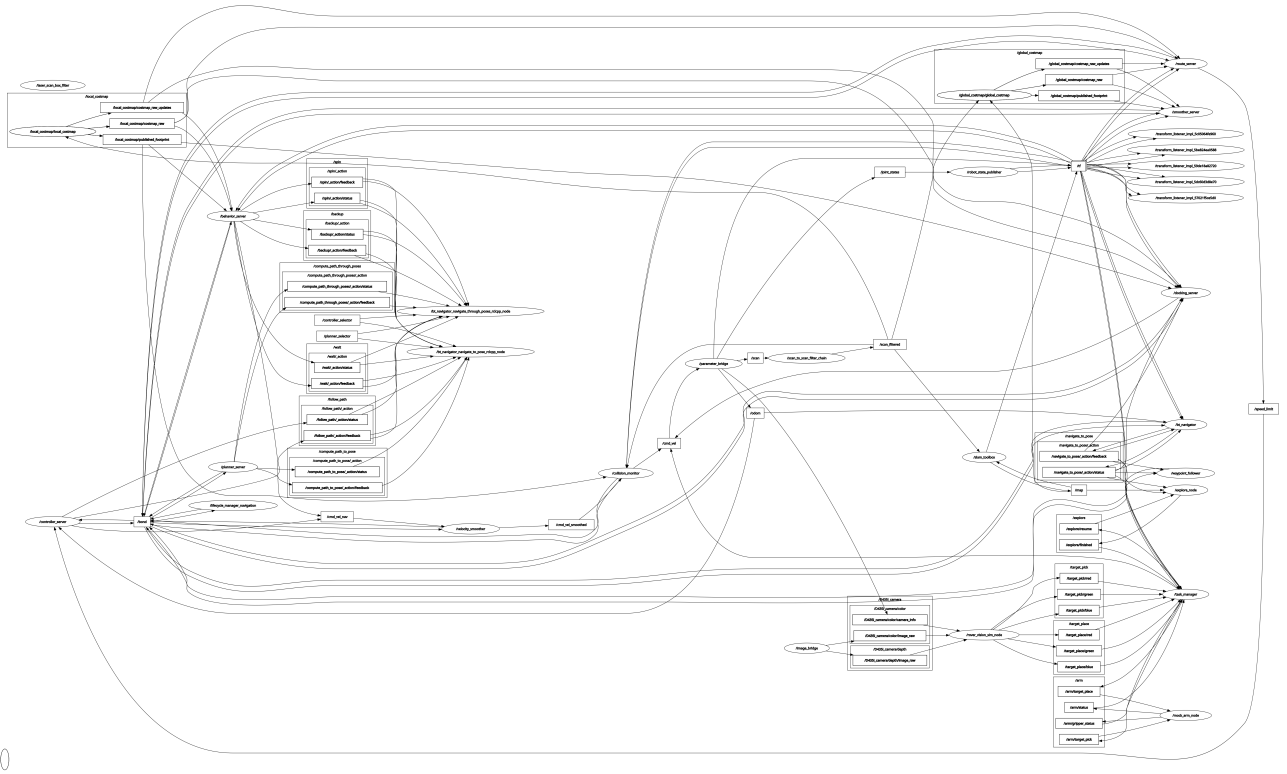


Fig. 18. RQT Graph visualizing active ROS 2 nodes, topic routing, and communication topology during autonomous operation.

As illustrated in the RQT graph, the runtime node topology strictly adheres to the proposed modular design:

- **Perception and Vision Link:** In the bottom right quadrant, the `d435_camera` node is actively publishing RGB and depth streams (e.g., `/d435_camera/color/image_raw`). This streams directly into the vision processing pipeline, aligning with the Intel RealSense D435 subsystem defined in the hardware block diagram and the `rover_vision_node` in the software architecture.
- **Navigation and Mapping Link:** The center and left sections of the graph are dominated by the `slam_toolbox` and Nav2 stack components (including `controller_server`, `planner_server`, and `bt_navigator`). These nodes ingest the `/scan` data originating from the RPLidar A2M12 driver. The dense topic routing here, culminating in the `/cmd_vel` output to the Leo Rover base, perfectly mirrors the mapping and trajectory planning logic established in Figure 16.
- **Central Orchestration and Actuation Link:** The `task_manager` node acts as the concrete realization of the Central Control State Machine (FSM). It interfaces bidirectionally with the `explore_node` for frontier-based navigation and coordinates with the `mycobot_node` to trigger grasping operations based on validated target coordinates.

The active connections verify that the high-bandwidth sensor data and critical control commands are isolated and routed correctly without causing network congestion. The asynchronous topic, service, and action communications shown here guarantee the deterministic execution of the autonomous mission flow, fully satisfying the requirements for decentralized, modular software control.

5.5 Github Repository Link

Link to Team Github Repository: https://github.com/longchuanYoRHa/Group12_team_space.git

6 Requirements Analysis and Verification

This section analyzes whether the design meets the course requirements using the Requirements Verification Matrix (RVM). Table 21 lists each shall-statement, its corresponding verifiable success criteria, and the analysis and verification plan based on the current mechanical, electrical, and software design.

Table 21. Requirements Verification Matrix

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
Operation						
1	The robot shall operate within the Test Environment.	The design of the robot utilises the Leo Rover 1.8 platform which has been proven to operate effectively in the lab environment. The mechanical design has not considered dynamic forces due to restricted movement speed and level area. No consideration for extreme conditions (heat, water ingress, etc.) necessary	Demonstration: During testing, when the robot is operating autonomously, a record is kept of any instance where ability to perform its function are impaired or impeded by environmental features of the test area. Video recorded.	Robot operation is not impeded by factors of the testing environment.	All	Pass: 0 Instances of impairment/impedance. Marginal: Minorly impaired/impeeded but still correctly performs function. Fail: Robot is unable to perform function.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
2	The robot shall be capable of restarting the operation.	The exploration workflow supports full restart after interruption. When the system is rebooted or halted, SLAM resets the map, the bringup node relaunched all navigation components, and the autonomous workflow begins again from the initial state. The robot can therefore resume operation reliably without manual reconfiguration.	Demonstration: The robot in a functioning state. Following power loss or some command to that effect the robot is brought back to a functioning state and resume operation.	Robot can be shown to restart the operation after interruption/from off state.	AB, QS	Pass: The robot can be restarted and brought back to a fully functioning state. Marginal: The robot can be brought back to a partially functioning state. Fail: The robot cannot be brought back to a fully functioning state.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
2.1	The robot shall achieve full operation in <5 minutes.	A timer is initiated at program start to monitor the 5-minute operational window. If available time becomes insufficient, the system safely increases navigation speed within the configured velocity limits, allowing SLAM initialization, exploration startup, and autonomous movement to complete within the required time-frame	Test: The robot is in a functioning state. For 3+ tests the robot is powered down to then be brought back to a functioning state. The time it takes to do this is recorded.	During testing the robot can be restarted and fully operational withing the 5 minute window.	AB, ZP	Pass: The times recorded are all less than 5 minuites. Marginal: <50% of the times are less than 5 minuites. Fail: None of the times are less than 5 minuites.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
3	All hardware shall be mounted on the robot platform.	The payload sled design allows for all components to be mounted to the robot securely and barring improper use should remain attached.	Demonstration: Each nessesary component is mounted onto the robotic platform and secured in place for the duration of a single cycle of normal operation. Video recorded.	There is no un-mounted hardware on the robot and this ramains true for the duration of operation.	AB, ZP	Pass: All harware remains mounted for the full duration of the demonstration. Marginal: All hardware is initially mounted and <50% remains mounted for the duration of the demonstartaion. Fail: Hardware is not mounted or >50% remains mounted for the duration of the
Navigation and planning						

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
4	The robot shall navigate autonomously.	Based on the hardware restrictions, and the support documents from Leo Rover. Currently the system integrated SLAM Toolbox with Nav2, forming a navigation stack that supports real-time mapping, localization, and autonomous motion planning. In addition, by incorporating the exploration node provided in the m-explore-ros2 package the robot is capable of fully autonomous map exploration. The LiDAR positioning on the payload sled allows 200 degrees of unobstructed sensing which has proven to be sufficient in preliminary testing.	Test: During testing the robot is placed in the test environment with known map, The robot is instructed to reach a target location using its autonomous navigation system. The robot's behaviour and path execution are recorded.	The robot successfully navigates the environment without external influence	SH	Pass: Robot reaches the target autonomously with minimum error. Marginal: Robot reaches the target with noticeable error Fail: Robot cannot reach the target autonomously.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
4.1	The robot shall avoid obstacles and unintended collisions.	Nav2 avoids obstacles by updating costmaps from LiDAR data, inflating obstacles based on the robot footprint, and selecting only collision-free trajectories through real-time trajectory simulation and controller evaluation.	Demonstration: During autonomous navigation the robot operates in the test environment containing walls, obstacles, coloured bins or boxes. Any instance of collision or contact with objects is recorded while the robot attempts to navigate. Video recorded.	During operation the robot does not impact the test environment wall, obstacles, coloured bins or boxes unintentionally.	SH, AB	Pass: No collisions occur during navigation. Marginal: Minor contact occurs but the robot continues functioning. Fail: Robot collides with obstacles and cannot continue operation.
4.2	The robot shall be capable of receiving commands from external controls.	The robot accepts external interrupts through ROS2 action and service interfaces. These allow higher-level controllers to issue navigation or manipulation commands, enabling safe task interruption and immediate behavioral reconfiguration during operation.	Test: External commands are sent to the robot control interface (e.g. ROS topic, keyboard command, or remote interface). The robot response to the command is recorded and verified. Video recorded.	The robot responds accordingly to commands issued from external controls	SH, QS	Pass: Robot responds correctly to commands. Marginal: Delayed or inconsistent response but command still executed. Fail: Robot does not respond to commands.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
4.4	Maximum linear velocity shall be limited to ≤ 250 mm/s.	Speed limits are enforced by configuring maximum linear and angular velocities in the Nav2 controller parameters, ensuring all generated trajectories and issued /cmd_vel commands remain within required safety constraints.	Test: The robot is commanded to navigate to certain position while its speed is measured using odometry or recorded motion data. The measured speed is measured from odometry or recorded motion data and verified against the requirement.	The robot does not exceed 250 mm/s	SH, AB	Pass: Robot velocity never exceeds 250 mm/s. Marginal: Robot slightly exceeds limit for short duration. Fail: Robot consistently exceeds velocity limit.
4.5	Maximum angular velocity shall be $<60^\circ/\text{s}$.	Angular velocity is constrained by setting the Nav2 controller's max_vel_theta parameter, ensuring all planned motions respect the required rotational speed limit and preventing unsafe or excessive turning during navigation.	Test: The robot is commanded to to navigate to certain position. Angular speed is measured from odometry or recorded motion data and verified against the requirement.	The robot does not exceed $60^\circ/\text{s}$	SH, AB	Pass: Angular velocity remains within limit. Marginal: Slight temporary exceedance of limit. Fail: Angular velocity consistently exceeds limit.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
5	The robot shall locally generate paths for navigation	Nav2 computes all global and local paths onboard using SLAM-generated maps and live sensor data, ensuring fully local planning without dependence on external computation or preloaded routes.	Demonstration: The robot is commanded to navigate to a target position while the local path planning process is monitored using Rviz. The locally generated path is recorded during operation. Video recorded.	The process of path planning is shown to be running locally.	SH, AB	Pass: Local path planning is shown to generate valid navigation paths. Marginal: Path planning functions inconsistently but navigation still occurs. Fail: Robot cannot generate paths locally.
5.1	Path planning shall use data from sensor data not pre-programmed global map	SLAM Toolbox builds the map in real time from LiDAR inputs, and Nav2 plans paths using these evolving costmaps, ensuring navigation relies solely on live sensor observations rather than predefined maps.	Demonstration: The robot performs navigation and mapping using live sensor data. Changes in the environment are introduced and the robot adjusts its planned path accordingly. Video recorded.	Sensor data is shown to be driving decision making of the path planning.	SH, AB	Pass: Sensor data actively drives path planning decisions. Marginal: Sensor data partially influences path planning. Fail: Robot relies only on pre-programmed map and does not react to sensor data.
Vision and perception						

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
6	The robot shall locate the Coloured Objects.	The system fuses 2D bounding boxes from YOLOv8/v11 with aligned depth frames from the RealSense D435i. This allows for the extraction of spatial ROI (Region of Interest) corresponding to specific objects.	Demonstration: The robot and the coloured objects are placed into the test environment. The robot is shown to identify the coloured objects relative to its own position. Various distances and positions are demonstrated. Video	The robot correctly locates the coloured objects within 3D space	QS, AB	Pass: The robot correctly locates the coloured objects at a range of distances. Marginal: The robot correctly/intermittently locates the coloured objects at limited ranges. Fail: The robot cannot locate the coloured objects.
6.1	The robot shall determine the 3d coordinates (x,y,z) of the target relative to the camera.	Using the camera's intrinsic matrix, the system deprojects pixel coordinates (u, v) and filtered depth values d into 3D camera coordinates (x, y, z). PCL statistical filtering removes outliers to ensure coordinate precision within $\pm 5\text{mm}$.	Test: The ability of the robot to provide the XYZ coordinates of the target is tested repeatedly at various distances in lab conditions. The results are compared to a alternatively measured value for accuracy. The	The robot accurately (within a grasp capable margin) provides the relative XYZ coordinates relative to the camera.	QS, AB	Pass: The accuracy of the XYZ coordinates are adequate for successful manipulation. Marginal: The robot provided coordinates but they are not adequate for successful manipulation. Fail: The robot fails to provide XYZ coordinates.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
6.2	The robot shall locate coloured blocks of size < 20 x 20 x 20mm	The RealSense D435i provides sufficient point cloud density at the manipulation range. The YOLO input resolution (e.g., 640x640) ensures that 20mm objects occupy a distinguishable pixel area for reliable detection.	Test: The blocks shall be measured and their size recorded. they will then be used as coloured objects in requirement 6.	Coloured blocks of size 20x20x20mm shall be used in the coloured object testing to measure success.	QS, AB	Pass: The robot correctly locates the coloured blocks at a range of distances. Marginal: The robot correctly/intermitently locates the coloured blocks at limited ranges. Fail: The robot does not locate the coloured blocks.
7	The robot shall classify the Coloured Objects	Classification is performed by the YOLOv8/v11 detection head. The model is optimized via OpenVINO and deployed on the NUC's Iris Xe iGPU to achieve real-time inference (>30 FPS) with high confidence scores.	Demonstration: The robot and the coloured objects are placed into the test environment. The robot is shown to classify the coloured objects. Various distances and positions are demonstrated. Video	The robot correctly classifies the coloured objects.	QS, AB	Pass: The robot correctly classifies the coloured objects at a range of distances. Marginal: The robot correctly/intermitently classifies the coloured objects at limited ranges. Fail: The robot cannot classify the coloured objects.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
7.1	The robot shall classify coloured blocks of size 20 x 20 x 20mm	The neural network is trained on a dataset containing small objects. Feature pyramid networks (FPN) within the YOLO architecture enhance the model's ability to extract semantic features from small (20mm) targets.	Test: The blocks shall be measured and their size recorded. they will then be used as coloured objects in requirement 7.	Coloured blocks of size 20x20x20mm shall be used in the coloured object testing to measure success.	QS, AB	Pass: The robot correctly classifies the coloured blocks at a range of distances. Marginal: The robot correctly/intermitently classifies the coloured blocks at limited ranges. Fail: The robot cannot classify the coloured blocks.
7.2	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	The training pipeline includes aggressive HSV augmentation (Hue, Saturation, Value jitter) and random brightness/contrast adjustments. This forces the model to learn colour invariance, ensuring robustness against shadows and lab lighting shifts.	Test: In lab conditions and in the test environment the ability of the robot to correctly locate and classify the coloured objects is tested at variable distances, under several different repeatable lighting conditions likely to be encountered.	The robot can be shown to correctly identify the bins colour under the natural lighting conditions in the lab.	QS, AB	Pass: The robot can tolerate a range of lighting conditions, including shadows. Marginal: The robot can tolerate some of the tested lighting conditions. Fail: The robot cannot tolerate lighting variation.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
7.3	The robot shall locate & classify at a distance >75mm	The RealSense D435i is configured to operate effectively from its minimum sensing distance (~0.28m) outwards. For distances >75mm (specifically in the arm's workspace), the depth accuracy is optimal for grasping planning.	Demonstration: The robot shall be placed into the test environment, a coloured object shall be placed >75mm away. The robot will attempt to locate and classify the coloured object. Video recorded.	The robot provides accurate positional data and correctly identifies the objects colour whilst positioned adjacent to the coloured objects mounting box.	QS, AB	Pass: The robot succeeds in locating and classifying the object. Marginal: The robot succeeds intermitantly in locating and classifying the object. Fail: The robot does not succeed.
8	The robot shall locate & classify the Coloured Bins.	Bins are detected as distinct classes (e.g., class: red_bin). Their 3D centroids are transformed via the ROS TF Tree into the global map frame, allowing the navigation stack to plan approach paths for depositing objects.	Demonstration: The robot shows that it can sucessfully locate and classify the coloured bins for a range of lighing conditions and distances.	The robot provides accurate positional data and correctly identifies the bin's colour during	QS, AB	Pass: The robot succeeds in locating and classifying the bins for all likely test conditions Marginal: The robot succeeds intermitantly in locating and classifying the bins. Fail: The robot does not succeed.
Object manipulation						

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
9	The robot shall manipulate the Coloured Objects	The manipulator subsystem transforms its pose into the manipulator base coordinate system via a transformation tree to execute a deterministic grasping process. The manipulator controller strictly enforces joint limits and positional accuracy, while the motion planner generates collision-free joint trajectories.	Demonstration: In lab and test conditions the robot is placed adjacent to the coloured object. The robot attempts to manipulate the coloured object in the desired way for a variety of distances and orientations.	The robot successfully manipulates the Coloured objects in the desired way.	NL,ZP	Pass: The robot successfully manipulates the coloured objects in the desired way for a variety of distances and orientations. Marginal: The robot manipulates the coloured objects but only for some of the distances and orientations. Fail: The robot fails to manipulate the coloured objects and/or loses control of the object.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
9.2	The minimum size of the Coloured Objects to be manipulated shall be 20 x 20 x 20mm.	The manipulator's repeatability accuracy (± 0.5 mm), gripper opening range (20-45 mm), gripping force, and control precision (positioning accuracy ± 1 mm) meet the requirements for grasping and transporting objects of this minimum size.	Analysis: The coloured objects are measured and the results recorded. They are then used in the same demonstration as requirement 9.	The robot successfully manipulates objects of the size specified.	NL, ZP	Pass: The robot successfully manipulates the specified coloured objects in the desired way for a variety of distances and orientations. Marginal: The robot manipulates the specified coloured objects but only for some of the distances and orientations. Fail: The robot fails to manipulate the coloured objects and/or loses control of the object.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
9.3	The robot shall manipulate at a height of 200mm above the floor	The manipulator's kinematics and joint limits enable the end effector to perform approach, grasp, and place motions at 200 mm. Visual depth perception provides pose correction for final alignment.	Demonstration: In lab and test conditions the robot is placed adjacent to the coloured object. The robot attempts to manipulate the coloured object when it is placed on a plinth 200mm above the floor. Video recorded.	The robot is shown to manipulate objects on top of the coloured block plinth and at the same height as the coloured deposit bins.	NL, ZP	Pass: The robot successfully manipulates the coloured objects in the desired way for a variety of distances and orientations. Marginal: The robot manipulates the coloured objects but only for some of the distances and orientations. Fail: The robot fails to manipulate the coloured objects and/or loses control of the object.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
9.4	The maximum weight shall be 150g	The payload capacities of the myCobot 280 Pi and gripper are 250g and 150g respectively. The gripper itself weighs 110g. Considering the weight of the building blocks, the upper limit selected is 50g.	Analysis: The coloured objects are weighed and the results recorded. They are then used in the same demonstration as requirement 9.	When calculated, none of the coloured blocks shall weigh more than 150g.,	NL, ZP	Pass: The robot successfully manipulates the coloured objects in the desired way for a variety of distances and orientations. Marginal: The robot manipulates the coloured objects but only for some of the distances and orientations. Fail: The robot fails to manipulate the coloured objects and/or loses control of the object.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
10	The robot shall transport the Coloured Objects.	The navigation stack (SLAM Toolbox + Nav2) generates collision-safe paths under local map and obstacle expansion constraints. The vehicle controller ensures linear and angular velocity limits (≤ 250 mm/s, $\leq 60^\circ$ /s) to maintain transport stability. After grasping, the robotic arm retains object in transport pose until approaching the drop point for precise placement.	Demonstration: In lab and test conditions the robot retains the coloured object. The robot attempts to move around the test environment while sucesfully retaining the coloured object. Video recorded.	The robot will successfully carry coloured objects between given locations.	NL, ZP, SH	Pass: The robot successfully retains the coloured objects in the desired way. Marginal: The robot retains the coloured objects but only for a limited time. Fail: The robot fails to retain the coloured objects and/or looses control of the object.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
10.1	The robot shall transport a minimum of 1 Coloured Object.	Preliminary testing has shown that the gripper can successfully grasp and hold a coloured object, and retain it whilst in motion. This is how the design shall transport a single coloured block. If required a 3 compartment coloured object container can be mounted to the existing payload sled to enable transport of multiple objects.	Demonstration: As above.	The robot will show it can transport a single coloured object inside the test environment.	NL, ZP, SH	Pass: The robot successfully retains the coloured object in the desired way. Marginal: The robot retains the coloured object but only for a limited time. Fail: The robot fails to retain the coloured object and/or loses control of the object.
11	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	The system will automatically relocate and retry within the permitted number of retries. All events (success/failure, retry count, timestamp) are logged to demonstrate the repeatability and reliability of three consecutive successful attempts.	Test: In lab and test conditions the robot retains the coloured object and is placed adjacent to the bins. The robot then attempts to deposit the coloured object into the bin for 3+ successive trials. Pass/failure is recorded.	For 3+ successive trials the robot will deposit the Coloured object into the correct bin.	NL, ZP	Pass: The first 3 successive trials are passed. Marginal: The robot achieves 3+ successive trials, but the completion rate is >100% Fail: The robot cannot complete 3 successive trials.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
Safety						

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
12	The robot shall be capable of being manually handled.	The mechanical design and positioning of components do not preclude the ability to safely handle the robot. Concideration has been given to not impede existing lifting points and to lifting strategy.	Demonstration: Prior to operational testing the robot is physically picked up by an individual from the lab bench and safely placed into the test area. Video recorded.	The robot shall be shown to be safely picked up and placed into the test environment.	NL, ZP	Pass: There is no difficulty in handling the robot and good lifting technique is maintained throughout. The robot remains fully assembled during the handling and no injurys occur. Marginal: There is some difficulty handling the robot. Adequate lifing technique is maintained. The robot remains fully assembled and no injurys occur. Fail: There is difficlty handling the robot. Poor lifting tequenique is demonstrated. The robot does not remain fully assembled or an injury occurs.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
12.1	The combined weight of the robot and the payload shall be <20kg.	The combined weight of all known components is 2.36 kg, the selected battery weighs 600g each (x2) and the payload sled is expected to add approximately 300g. The rover base contributes 6.50 kg, bringing the total estimated weight to about 10.36 kg. This remains below the target weight constraint, ensuring compliance with performance requirements.	Analysis: The weights of all the components is summed from the data sheets with the calculated/measured weights of the additional components. The values are recorded and compared to the 20kg benchmark.	When calculated the sum of weights of all the components and the payload shall be less than 20kg	NL, AB	Pass: <20kg Marginal: 20kg Fail: >20kg
13	The robot shall be able to be stopped in the case of unexpected behaviour.	The Leo Rover and C300 have physical power buttons that cut power when pushed, and will serve as emergency shut-off. This is deemed acceptable risk given the controlled environment.	Demonstration: During normal operation or unexpected behavior in the test area, a command shall be given and the robot shown to be stopped by some method. Video	Emergency shutdown procedure works as intended.	ZP	Pass: After the command issued the robot immediately ceases its activity. Marginal: After the command issued the robot ceases its activity in >20s. Fail: The robot does not cease its activity.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
13.1	The robot shall be capable of being issued a stop command wirelessly.	The robot can be stopped wirelessly via Bluetooth or an existing SSH connection. A remote operator issues a software interrupt that cancels all active ROS2 actions and commands the bringup node to publish a zero-velocity override, ensuring rapid and reliable termination of ongoing navigation or manipulation tasks.	Demonstration: During normal operation or unexpected behavior in the test area, a wireless command shall be given and the robot shown to be stopped. Video Recorded.	The robot will show the ability to be stopped remotely.	SH, QS	Pass: After the command issued the robot immediately ceases its activity. Marginal: After the command issued the robot ceases its activity in >20s. Fail: The robot does not cease its activity.
13.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	The addition of red tape (or some other visual indicator) to the Leo Rover, and C300 power buttons shall provide visual indication of power cut-off	Inspection: Upon inspection a button should be present that is clearly labeled as emergency shutdown.	Emergency stop button is present, indicated, and accessible	ZP, SH	Pass: Button is present, is clearly labeled and is reachable without performnig any additional steps or requirning specialist knowledge. Marginal: Button is present but not nessesarily clearly labeled. Fail: Button is not present.

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Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
Power						

Continued on next page...

Table 21. Requirements Verification Matrix (Continued)

Req #	Shall Statement	Analysis of design	Verification Method	Verification of Success Criteria	Resp.	Grade
14	Electrical design and maintenance shall comply with lab risk assessment	The twin power supply delivers the required wattages to the NUC, peripherals, manipulator and gripper with ample margin for the demonstration, while the Leo Rover base remains powered by its OEM battery. All power components and OEM adapters/cables are selected from reputable suppliers and manufacturers who perform their own testing and quality inspection, minimizing the risk of short circuits, voltage transients, or electrical noise, and operating thresholds follow the manufacturers' specifications. During the 20-minute mission the rover is expected to complete the specified tasks without depleting the battery, operate without short circuits or electrical damage to components, and maintain full performance with no degradation caused by insufficient power.	Inspection: Upon inspection the robot complies with each relevant item on the lab risk assesment. The results are recorded.	The power supply and batteries can be shown to comply with the risk assessment by design and when inspected.	AB, ZP	Pass: The Robot complies with all items on the lab risk assesment. Marginal: The robot complies with most, and it is unclear on the others. Fail: The robot fails to comply with one or more requirement.

7 Sustainability for Robotics Checklist

7.1 Sustainability Overview

This overview outlines the core sustainability strategies integrated into the design of the mobile manipulator. These considerations aim to ensure compliance with environmental standards while optimizing the system's operational lifecycle.

- 1. Materials** Materials are generally durable and repairable, with metal frames, PCR plastics, and long-life batteries prioritized where possible. Opportunities remain to increase recycled content, switch to lower-impact chemistries (e.g., LFP), and replace single-use resins with recyclable or bio-based alternatives.
- 2. Software** Software supports energy savings and reliability through features like auto-shutdown, adaptive sensor modes, and integrated BMS telemetry. Continued focus on lightweight, open-source tooling and low-code monitoring will improve operational efficiency and extend hardware life.
- 3. Energy** Energy performance is strong at the system level (efficient motors, ENERGY STAR components, high conversion battery tech), but larger image inputs and high-rate sensors increase consumption. Smart charging, regenerative braking, and deeper sleep modes offer clear gains for field deployments.
- 4. Waste** Waste reduction is enabled by modular design, long component lifetimes, and manufacturer take-back potential. Formalizing end-of-life recycling, repair programs, and spare-part availability will materially reduce e-waste.
- 5. Emissions** Emissions are low in operation (electric drive, no tailpipe emissions) and some suppliers use low-carbon manufacturing; packaging and supply-chain emissions remain the main levers. Supplier transparency and low-carbon logistics would strengthen the rover's climate profile.
- 6. Communications** Communications are designed for efficiency (bandwidth controls, voxel filtering, PD power negotiation), helping reduce transmission energy. Adding smarter scheduling and compression further lowers network and cloud energy costs.
- 7. Modularity** Modularity is a clear strength: standardized mounts, tool-less access, and interoperable connectors make upgrades and repairs straightforward. This design philosophy directly supports reuse, repair, and reduced material consumption.
- 8. Location and Deployment** Location/Placement considerations (portable batteries, rugged housings, local sourcing) make the rover adaptable to field work with lower transport impacts. Local procurement and reusable shipping would further cut lifecycle emissions.
- 9. Maintenance** Maintenance is enabled by diagnostics, modular parts, and accessible internals, which together extend service life and reduce replacements. Better documented maintenance procedures and predictive alerts will improve uptime and sustainability.
- 10. Repurposing** Repurposing options are strong: second-life batteries, donated sensors, and upcycling guides enable meaningful circularity. Formal second-life programs and public configuration logs would maximize reuse and social benefit.

7.2 Sustainability Checklist

Table 22. Sustainability Checklist

Component	Assessment & Values	Possible Improvements	UN SDG Met
1. Materials			
Leo Rover 1.8 (Base Platform)	Aluminum frame and plastic covers; IP66 watertight	Use recycled aluminum for internal frame components	SDG 12.2: Sustainable resource use via recycled aluminium
Leo Rover Battery Pack (Rover Power System)	Li-ion 5000 mAh; IP66 case	Transition to LFP chemistry for longer cycle life	SDG 12.4: Environmentally sound management of battery chemicals
Intel RealSense D435 (Vision Sensor)	Precision glass and semiconductor assembly	Use mounts made from recycled PETG filament	SDG 12.2: Efficient use of materials via recycled PETG
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	Brushless motor and optical glass components	Request a full RoHS report for internal parts	SDG 12.4: Responsible management of electronic components
Anker Prime 26K 300W (Payload Battery, 2 units)	ClimatePartner certified for measured and offset lifecycle emissions; recycled materials used in some cables	Publish a full bill of materials (BOM) including exact PCR plastic percentages	SDG 12.6: Encourage sustainability reporting and transparency

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Elephant Robots myCobot 280 Pi	Industrial-grade nylon and aluminium alloy construction with metal/steel internal gearing	Transition to bio-based filaments for non-structural plastic casings	SDG 12.2: Sustainable material use via bio-based filaments
Elephant Robotics Adaptive Gripper	Photosensitive resin and lightweight plastics with metal internal structural gears	Replace resin components with CNC aluminium for higher wear resistance	SDG 12.2: Efficient resource use via durable CNC aluminium parts
ASUS NUC 12 Pro (NUC12WSKi7)	PCR plastics used in chassis; responsible sourcing for tantalum, tin, tungsten, and gold	Increase PCR plastic content in internal fan housing to >50%	SDG 12.2: Resource efficiency via increased recycled content
System Wiring & Connectivity	TPE/PVC jacketing with copper shielding; PD-to-5.5mm barrel cables (12V & 20V) for high-wattage delivery	Transition to halogen-free (LSZH) cabling to reduce toxic off-gassing	SDG 12.4: Reduce hazardous emissions via halogen-free cabling
Payload Sled	3D printed PLA	Use recycled PLA filament	SDG 12.2: Sustainable resource use via recycled PLA
2. Software			

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Leo Rover 1.8 (Base Platform)	Ubuntu/ROS-based; open-source ready	Implement auto-shutdown for the Raspberry Pi during idle	SDG 7.3: Improve energy efficiency through idle auto-shutdown
Leo Rover Battery Pack (Rover Power System)	Internal BMS for short-circuit and safety protection	Integrate BMS data into the ROS dashboard for health monitoring	SDG 9.1: Strengthen infrastructure reliability via integrated BMS monitoring
Intel RealSense D435 (Vision Sensor)	Global shutter; RealSense SDK 2.0	Use depth-only mode when color is not required	SDG 7.3: Reduce energy use by disabling RGB when unnecessary
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	16K sampling; 10 Hz scan rate	Dynamically adjust scan frequency based on vehicle speed	SDG 7.3: Energy efficiency via adaptive scan frequency
Anker Prime 26K 300W (Payload Battery, 2 units)	ActiveShield 4.0 performs extensive temperature checks to prevent overheating	Integrate battery health and cycle-count data into the companion app for user visibility	SDG 9.4: Sustainable upgrades via transparent battery health analytics
Elephant Robots myCobot 280 Pi	Native Ubuntu MATE 20.04 supporting Python and ROS1/ROS2	Develop a low-code web interface for tablet-based energy monitoring	SDG 9.5: Support scientific research and innovation via open-source software
Elephant Robotics Adaptive Gripper	PWM control via Python/C++ APIs	Implement haptic feedback loops to detect grip success without vision	SDG 9.5: Advance research through haptic sensing integration

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
ASUS NUC 12 Pro (NUC12WSKi7)	Intel vPro for remote management; ActiveShield thermal monitoring with automated alerts	Add a native carbon footprint tracker to the NUC Pro software suite	SDG 9.4: Sustainable upgrades via carbon tracking software features
System Wiring & Connectivity	Integrated E ² Marker chips in PD cables to negotiate power profiles safely	Use 'Cable ID' tags in the NUC OS to monitor power draw per line	SDG 7.3: Improve energy efficiency via smart power negotiation
Payload Sled	Created using SOLIDWORKS Student Edition 2024	Use generative design practices to minimize material usage	SDG 12.2: Reduce material use via generative design
3. Energy			
Leo Rover 1.8 (Base Platform)	4 × in ³ hub DC motors; 73.2:1 gearboxes	Optimize motor controllers for regenerative braking	SDG 7.3: Energy efficiency via regenerative braking
Leo Rover Battery Pack (Rover Power System)	4 hours driving / 8 hours streaming per charge	Use smart chargers that sync with renewable energy	SDG 7.2: Increase renewable energy share
Intel RealSense D435 (Vision Sensor)	USB-C powered; peak consumption 5 W	Power down the IR projector when the rover is stationary	SDG 7.3: Energy efficiency via low-power vision

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	Low wattage: 2.25–3 W peak consumption	Use motor speed control to save power when idle	SDG 7.3: Energy efficiency via low□wattage operation
Anker Prime 26K 300W (Payload Battery, 2 units)	300 W max output; 250 W dual□cable recharging; 97% peak conversion efficiency	Optimize idle standby power consumption to reduce energy waste	SDG 7.3: Energy efficiency via advanced GaN technology
Elephant Robots myCobot 280 Pi	12V/5A (60 W) input with high□efficiency servos	Implement a deep□sleep mode for the Pi when the arm is idle	SDG 7.3: Energy efficiency in automated desktop systems
Elephant Robotics Adaptive Gripper	Low power consumption (5V/1A) drawn from the arm’s end□effector port	Add a torque□limit mode to reduce power once a firm grip is achieved	SDG 7.3: Energy efficiency via low□voltage actuator control
ASUS NUC 12 Pro (NUC12WSKi7)	ENERGY STAR certified; ultra□low idle power (6–10 W) with 120 W adapter	Enable Eco□Mode in BIOS as the default factory setting	SDG 7.3: Energy efficiency via optimized system settings
System Wiring & Connectivity	USB 3.1 PD efficiency minimizes conversion loss; Ethernet control reduces signal noise energy waste	Shorten power cables to exact lengths to reduce copper resistance and heat	SDG 7.3: Energy efficiency via low□loss power distribution

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Payload Sled	Printed using UK grid electricity	Power the printer from exclusively green energy sources	SDG 7.2: Increase renewable energy share
4. Waste			
Leo Rover 1.8 (Base Platform)	Non-pneumatic tires to prevent rubber air leaks	Partner with Leo Rover for a hardware take-back program	SDG 12.5: Substantially reduce waste generation
Leo Rover Battery Pack (Rover Power System)	PCM circuit prevents over-drain damage	Create a disposal roadmap with certified recyclers	SDG 12.4: Environmentally sound management of battery waste
Intel RealSense D435 (Vision Sensor)	Intel e-waste recycling and trade-in programs	Use the manufacturer's trade-in program for upgrades	SDG 12.5: Reduce waste through corporate recycling initiatives
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	5-year operational life reduces e-waste	Ensure the optical window is cleaned rather than replaced	SDG 12.5: Reduce waste via long-life components
Anker Prime 26K 300W (Payload Battery, 2 units)	24-month warranty and high-density cells designed for longevity	Launch a formal take-back and repair program for individual battery cells	SDG 12.5: Reduce waste through durable design and repairability

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Elephant Robots myCobot 280 Pi	Modular design allows quick disassembly of individual joints	Offer a joint refurbishment service to extend lifespan	SDG 12.5: Reduce waste through field-repairable components
Elephant Robotics Adaptive Gripper	Replaceable finger pads and modular screw-on assembly	Provide 3D-printable STL files for replacement pads	SDG 12.5: Reduce waste via component-level repairs
ASUS NUC 12 Pro (NUC12WSKi7)	Modular design with dual SO-DIMM and M.2 slots for upgrades	Launch a mainboard swap program to upgrade CPUs without replacing the chassis	SDG 12.5: Reduce e-waste through easy dismantling and repair
System Wiring & Connectivity	Fixed Lidar wiring reduces connector wear; standard Ethernet and Type-C cables are replaceable	Use braided sleeving to protect cables from mechanical abrasion	SDG 12.5: Prevent waste by using interchangeable standard leads
Payload Sled	End-of-life process currently unknown; assumed recycling of support structures	Define and implement end-of-life recycling processes	SDG 12.5: Substantially reduce waste generation
5. Emissions			

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Leo Rover 1.8 (Base Platform)	Zero tailpipe emissions; electric drive	Use biodegradable lubricants for moving parts	SDG 11.6: Reduce local environmental impact in urban areas
Leo Rover Battery Pack (Rover Power System)	Zero gaseous discharge; heat is the only emission	Improve thermal casing to reduce efficiency loss	SDG 7.3: Reduce energy loss and lifecycle emissions via improved efficiency
Intel RealSense D435 (Vision Sensor)	Manufacturing powered by 100% renewable electricity	Source components from vendors with carbon-neutral shipping	SDG 13.2: Support climate mitigation through low-carbon production
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	Class 1 laser safety standard (IEC 60825)	Verify calibration to prevent laser scattering	SDG 11.6: Reduce environmental and health impacts through proper calibration
Anker Prime 26K 300W (Payload Battery, 2 units)	ClimatePartner certified with lower carbon emissions versus traditional manufacturing	Shift to fully plastic-free, 100% recyclable regional packaging	SDG 13.2: Contribute to climate mitigation via lower product carbon footprint
Elephant Robots myCobot 280 Pi	Collaborative cobot design with collision detection	Conduct a lifecycle assessment for regional shipping footprints	SDG 11.6: Reduce local environmental impact via safer urban deployment
Elephant Robotics Adaptive Gripper	Silent motor operation for non-disruptive environments	Verify RoHS compliance for all internal soldering	SDG 11.6: Reduce indoor noise and environmental impacts

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
ASUS NUC 12 Pro (NUC12WSKi7)	ClimatePartner certified; ASUS validated SBTi targets for supplier reductions	Transition to 100% plastic-free, soy-ink packaging for global distribution	SDG 13.2: Support emissions mitigation through supplier decarbonization
System Wiring & Connectivity	Zero gaseous discharge; STP Ethernet prevents EMI leakage	Use biodegradable cable ties instead of plastic zip ties	SDG 11.6: Reduce environmental impact via cleaner signal transmission
Payload Sled	Possible microplastic generation since PLA is not naturally biodegradable	Ensure industrial processing to break down polymers at end of life	SDG 12.4: Environmentally sound management of chemicals and waste
6. Communications			
Leo Rover 1.8 (Base Platform)	2.4 GHz Wi-Fi (100 m range); 5 MP video	Use data compression to reduce Wi-Fi energy use	SDG 7.3: Energy efficiency via reduced data transmission
Leo Rover Battery Pack (Rover Power System)	Real-time voltage monitoring in the main UI	Implement low-battery alerts to prevent cell damage	SDG 12.8: Provide information for sustainable lifestyles
Intel RealSense D435 (Vision Sensor)	USB 3.0 bandwidth for dense point clouds	Perform voxel filtering to reduce data load	SDG 7.3: Energy efficiency via reduced data processing

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	UART/USB serial interface for efficient data	Use hardware filtering to minimize CPU load	SDG 7.3: Energy efficiency via reduced processing
Anker Prime 26K 300W (Payload Battery, 2 units)	PD 3.1 support; Bluetooth app control for status and firmware updates	Add power□scheduling features to automate charging during off□peak hours	SDG 7.2: Increase renewable energy use via smart charging
Elephant Robots myCobot 280 Pi	4× USB, 2× micro□HDMI, and RJ45 Ethernet for IoT connectivity	Add Power□over□Ethernet (PoE) support to reduce cable clutter	SDG 9.c: Expand access to information and communications technology
Elephant Robotics Adaptive Gripper	Standard 4□pin interface for synchronized data and power	Integrate a digital ID chip so the arm auto□recognizes the gripper type	SDG 9.c: Improve access to interoperable information technology
ASUS NUC 12 Pro (NUC12WSKi7)	Dual Thunderbolt 4 ports, 2.5 Gb Ethernet, Wi□Fi 6E	Optimize Wake□on□LAN firmware to lower deep□sleep power states	SDG 7.3: Energy efficiency via optimized connectivity
System Wiring & Connectivity	USB 3.0/3.1 bandwidth for RealSense point clouds; 2.5 Gb Ethernet compatibility	Enable Ethernet Energy Efficient (EEE) mode for adapters	SDG 9.c: Access to high□speed information technology

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Payload Sled	Files transferred via Teams and stored on the cloud	Avoid duplicate files and unnecessary data transmission	SDG 7.3: Energy efficiency via reduced cloud usage
7. Modularity			
Leo Rover 1.8 (Base Platform)	Standard 18×15 mm hole grid for payloads	Design a universal mount that works across platforms	SDG 12.2: Efficient use of resources via interoperable mounts
Leo Rover Battery Pack (Rover Power System)	Standard 3-pin waterproof connector	Use housing that allows for cell-level replacement	SDG 12.2: Sustainable management via easy swapping
Intel RealSense D435 (Vision Sensor)	Standard 1/4-20 UNC and M3 threads for mounting	Use quick-release mounts between different robots	SDG 12.2: Resource efficiency via interoperability
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	Standard 4-pin interface; 76 mm footprint	Standardize mounting plates across A1, A2, and A3 models	SDG 12.2: Resource efficiency via standardized components
Anker Prime 26K 300W (Payload Battery, 2 units)	Pogo-pin interface allows seamless docking with the Anker charging base	Standardize the Pogo-pin layout for third-party compatibility	SDG 12.2: Resource efficiency via cross-device charging ecosystems

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Elephant Robots myCobot 280 Pi	LEGO-compatible base and end tip for easy integration	Standardize the gripper mount for 1:1 industrial tool swap-outs	SDG 12.2: Resource efficiency via cross-brand compatibility
Elephant Robotics Adaptive Gripper	Adaptive two-finger design capable of gripping varied shapes	Develop a quick-release latch to swap grippers without tools	SDG 12.2: Resource efficiency via multi-use tooling
ASUS NUC 12 Pro (NUC12WSKi7)	Tool-less chassis access with support for up to 64 GB RAM and Gen4 NVMe	Standardize internal expansion headers for universal sensor/IoT modules	SDG 12.2: Resource efficiency via modular internal bays
System Wiring & Connectivity	USB-to-Ethernet converters and PD-to-barrel adapters allow quick swaps	Standardize all power leads to a locking 5.5 mm barrel connector	SDG 12.2: Resource efficiency via interoperable connectors
Payload Sled	Interlocking and modular components	Redesign interfaces to reduce the number of steel fasteners required	SDG 12.2: Sustainable management of natural resources
8. Location/Placement			
Leo Rover 1.8 (Base Platform)	Designed for difficult terrains (Fukushima tested)	Ship in reusable crates rather than cardboard boxes	SDG 11.2: Support sustainable transport systems

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
Leo Rover Battery Pack (Rover Power System)	Portable; adheres to dangerous goods air rules	Charge via solar panels during field deployments	SDG 11.a: Strengthen regional environmental links via portable energy
Intel RealSense D435 (Vision Sensor)	Global supply chain with options for local purchasing	Purchase locally to minimize transport mileage	SDG 11.a: Support regional environmental links through local procurement
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	80,000 lux light resistance for outdoor use	Use a dust-proof cover for arid deployments	SDG 9.4: Upgrade technology to be more sustainable and resource-efficient
Anker Prime 26K 300W (Payload Battery, 2 units)	99.54 Wh capacity (TSA-approved <100 Wh) for air travel	Provide a weather-resistant carrying case for field use	SDG 11.a: Support global mobility via air-safe portable energy solutions
Elephant Robots myCobot 280 Pi	Lightweight 850 g profile and 280 mm working radius for desktop deployment	Design a battery-powered mobile base for off-grid field testing	SDG 11.a: Support regional links via portable research tools
Elephant Robotics Adaptive Gripper	Ultra-light 88 g weight to preserve payload capacity	Offer various stroke lengths for specific industrial tasks	SDG 11.a: Improve mobility via minimal weight impact

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
ASUS NUC 12 Pro (NUC12WSKi7)	Ultra-compact footprint with VESA mount compatibility	Offer a rugged, fanless industrial shell for dusty deployments	SDG 11.a: Support space-efficient urban deployment
System Wiring & Connectivity	1.2 m cable lengths provide slack for full manipulator reach and NUC mounting flexibility	Use a drag chain or cable snake to prevent tangling during 6-axis movement	SDG 11.a: Support mobile deployment via robust cable management
Payload Sled	Filament origin unknown	Source filament locally to reduce transport emissions	SDG 12.7: Support sustainable public procurement
9. Maintenance			
Leo Rover 1.8 (Base Platform)	Vibration-proof connectors for long-term use	Include internal humidity sensors to monitor seal integrity	SDG 9.1: Reliable and sustainable infrastructure through preventive maintenance
Leo Rover Battery Pack (Rover Power System)	Requires balanced charging protocols	Automate charge reminders in the rover software	SDG 9.1: Ensure resilient infrastructure via routine maintenance
Intel RealSense D435 (Vision Sensor)	Solid-state design with no moving parts	Use a sapphire glass shield to protect lenses	SDG 9.1: Reliable infrastructure through high mean time between failures

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	OPTMAG non-contact transmission technology	Check for debris in the rotation gap to prevent strain	SDG 9.4: Sustainable upgrades via non-contact transmission that reduce wear
Anker Prime 26K 300W (Payload Battery, 2 units)	Smart TFT color display provides real-time port health and battery temperature alerts	Include a user-accessible deep-discharge reset tool for battery calibration	SDG 9.4: Sustainable upgrades via smart diagnostics that improve reliability
Elephant Robots myCobot 280 Pi	Self-diagnostic firmware via MyStudio for motor calibration and health checks	Introduce auto-lubrication alerts based on joint rotation cycles	SDG 9.1: Reliable infrastructure through digital twins and predictive maintenance
Elephant Robotics Adaptive Gripper	External access to internal gears for manual cleaning and lubrication	Include a protective dust-boot sleeve to prevent debris in gear teeth	SDG 9.1: Reliable infrastructure through easy-access maintenance
ASUS NUC 12 Pro (NUC12WSKi7)	System health and temperature alerts via BIOS-level diagnostics and ASUS support software	Introduce a dust-sense notification to alert users when fans require cleaning	SDG 9.1: Reliable infrastructure through continuous monitoring

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
System Wiring & Connectivity	Vibration-proof Ethernet connectors and gold-plated Type-C contacts ensure reliable connectivity in high-movement robotics	Color-code 12V and 20V cables to prevent accidental high-voltage damage	SDG 9.4: Sustainable upgrades via high-durability connectors and safer wiring
Payload Sled	3D printer maintenance procedure currently unknown	Ensure the 3D printer is well maintained, clean, leveled, and firmware is up to date to reduce failed parts and filament waste	SDG 12.5: Reduce waste through improved maintenance
10. Repurposing			
Leo Rover 1.8 (Base Platform)	Payload capacity up to 5 kg for diverse tasks	Maintain a public log of hardware configurations for reuse	SDG 12.5: Reduce waste through prevention and reuse
Leo Rover Battery Pack (Rover Power System)	Stationary storage potential for second-life use	Use degraded batteries for low-draw IoT stations	SDG 7.b: Expand infrastructure for sustainable energy via second-life batteries
Intel RealSense D435 (Vision Sensor)	Used in AR/VR, security, and robotics education	Donate legacy sensors to local schools	SDG 4.4: Increase skills for employment through technology reuse

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
SLAMTEC RPLIDAR A2M12 (Lidar Sensor)	Reusable for vacuum robots or facility security	Use retired units for low-speed monitoring	SDG 4.4: Support skills development via repurposing
Anker Prime 26K 300W (Payload Battery, 2 units)	High output enables use as an emergency 300 W UPS for workstation backups	Offer a second-life trade-in program for off-grid educational tools	SDG 4.4: Build relevant skills for managing high-tech power systems
Elephant Robots myCobot 280 Pi	Open-source driver library allows retired units to be reused for automated tasks	Publish official upcycling guides for end-of-life hardware	SDG 4.4: Increase skills for second-life tech applications
Elephant Robotics Adaptive Gripper	Reusable across myCobot models (280 M5, Pi, and Jetson)	Create a mounting adapter for use with other open-source robot brands	SDG 4.4: Support cross-platform automation skills
ASUS NUC 12 Pro (NUC12WSKi7)	12-core performance enables retired units to serve as home assistant nodes or edge servers	Develop an official repurposing kit (fanless case and OS) for second-life IoT use	SDG 4.4: Build career-relevant skills in low-power server management

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Table 22. Sustainability Checklist (Continued)

Component	Assessment & Values	Possible Improvements	UN SDG Met
System Wiring & Connectivity	Universal Type-C PD cables can be repurposed for laptop charging or stationary IoT stations after rover retirement	Maintain a wiring schematic in the public log for future hardware reuse	SDG 4.4: Support skills for managing modular power systems
Payload Sled	End-of-life process currently unknown	Plan for possible reuse in future rover projects and define recycling pathways	SDG 12.5: Reduce waste through prevention and reuse

8 Cyber Security Considerations

The cyber security strategy for the robotic system is designed to ensure the integrity of the autonomous control logic and protect the system from unauthorized access. The following points have been considered:

8.1 Physical Security & Deterministic Network Interface Binding

The system's first line of defense is established through physical isolation and strict traffic routing logic:

- **Wired Point-to-Point Communication Isolation:**

The ASUS NUC and Leo Rover chassis are directly connected via a shielded Ethernet cable. This physical connection eliminates risks present in shared laboratory Wi-Fi environments, including Man-in-the-Middle (MITM) attacks, data sniffing, and deauthentication attacks. All traffic related to core robotic operations—such as motion control (e.g., `/cmd_vel`) and visual perception (e.g., depth map data)—is confined within this private physical link.

- **Deterministic Resolution of Multi-Network Interface Conflicts:**

To address communication conflicts in a multi-network environment (simultaneous Internet access for model downloads and an internal control network for operation), the team enforced binding of ROS2 discovery and data traffic to the Ethernet interface by modifying `FASTRTPS_DEFAULT_PROFILES_FILE`. This configuration prevents sensitive data, such as object grasping coordinates, from leaking into wireless networks. Additionally, disabling unnecessary network interface broadcasts reduces the overall attack surface.

8.2 Middleware Hardening via FastDDS XML

By implementing a dedicated FastDDS XML configuration, the system is hardened at the middleware level, ensuring the integrity of distributed computation:

- **Participant Discovery Filtering:**

XML-based bindings restrict the discovery scope of ROS2 domain participants. Only nodes with pre-authorized attributes are permitted to join the computational graph. This ensures that even if unauthorized devices gain physical access to the same local network, they cannot probe or hijack the robot's control flow via the standard ROS2 Simple Participant Discovery Protocol (SPDP).

- **Resource Quotas & Denial-of-Service (DoS) Resistance:**

Strict resource limits are configured for critical topics, including maximum sample counts, buffer size constraints, and history depth control. These measures effectively mitigate malicious topic flooding attacks that could otherwise exhaust the NUC's computational resources, thereby preserving the real-time performance and high availability of YOLO11-based visual inference under extreme network load.

8.3 Strategic Topic Routing & Multi-level Command Validation

At the application layer, the system introduces dynamic security mechanisms to ensure both the authenticity and physical safety of motion commands:

- **Gatekeeper Node Validation:** A “validation gateway” node is deployed between the vision processing node and actuator drivers. This node continuously verifies the X, Y, Z coordinates produced by the vision system, ensuring they remain within the robotic arm’s physical workspace limits. Any out-of-bound values (soft limit violations) or abnormal acceleration spikes are immediately intercepted, preventing mechanical damage caused by algorithmic noise or malicious command injection.

- **Heartbeat Monitoring & Fail-safe Mechanism:**

The routing logic incorporates bidirectional heartbeat monitoring. If a camera data stream interruption is detected or if actuator response times out, the system triggers a fail-safe mechanism. This immediately publishes zero-velocity commands and transitions the system into a safe lock state until manual intervention is performed.

8.4 Power Stability & Hardware-Level Availability Assurance

The integration of the Anker Prime 26K (250W/300W) portable power supply is critical for ensuring system availability:

- **Electrical Integrity & Ripple Suppression:** The power supply provides high-power regulated output (up to 140W per port), effectively resolving previous issues caused by multiple high-load devices (NUC + robotic arm) competing for current, which led to transient voltage drops.

- **Driver Failure Prevention Mechanism:**

Stable power delivery ensures that the NUC’s USB controller does not reset under high computational loads (e.g., 1280-resolution inference). This mitigates risks of driver crashes—such as device descriptor request failures—and prevents system-level denial-of-service conditions caused by hardware instability.

9 Updated Project Plan

The detailed project schedule, milestones, and task allocations are illustrated in the Gantt chart in Figure 19.

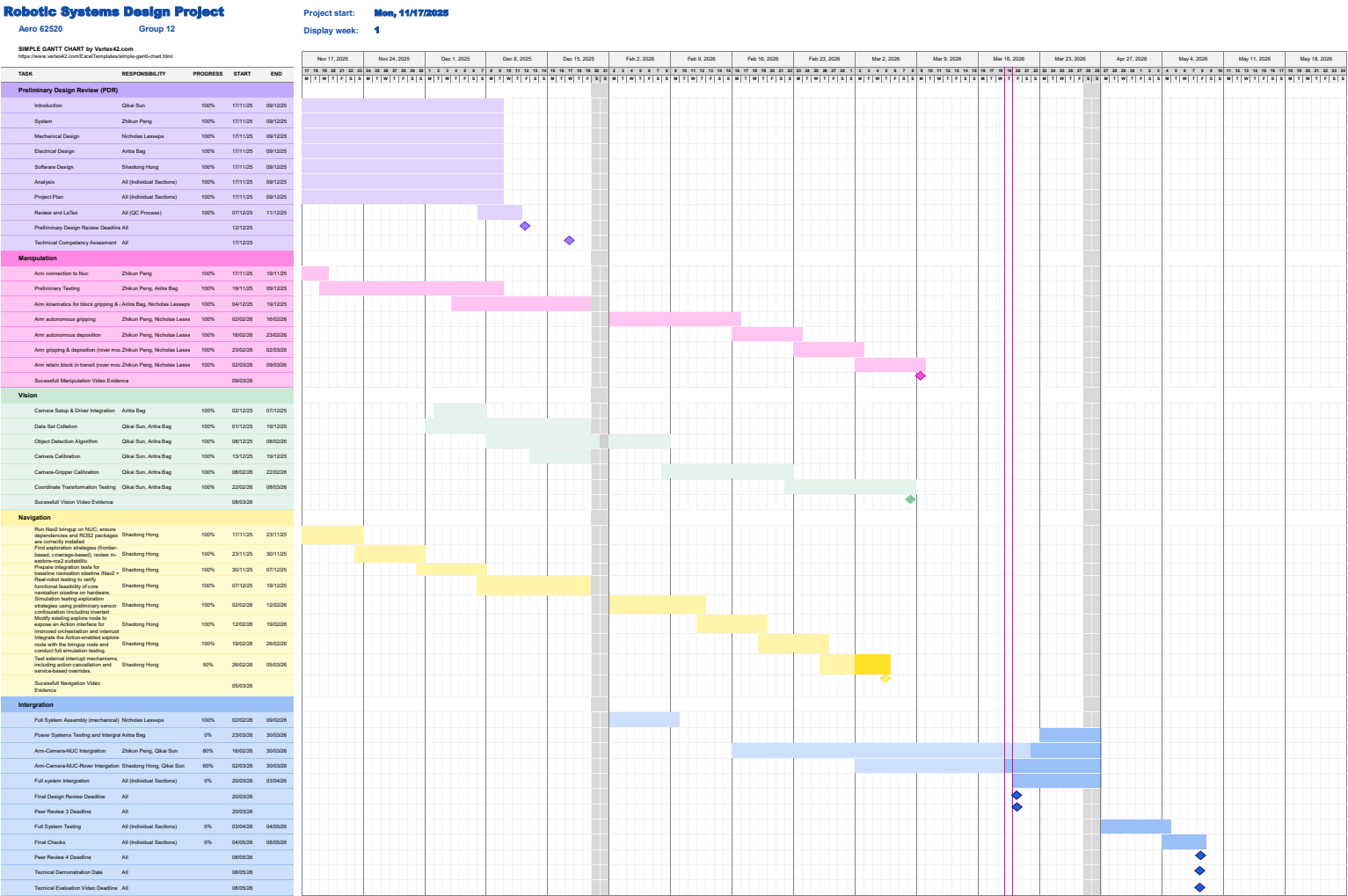


Fig. 19. Robotics Systems Design Project Gantt Chart

10 Conclusion

This Final Design Review (FDR) report comprehensively outlines the finalized design decisions, engineering rationale, and system architecture for the autonomous mobile manipulator. Building upon the foundation established in the Preliminary Design Review (PDR), the design has been significantly refined to address prior feedback and ensure strict compliance with the objectives specified in the Design Requirements Analysis.

Major modifications introduced in this phase include a highly optimized electrical architecture utilizing a dual Anker Prime 26K power supply. This transition drastically reduced the system's payload weight by over 70% while fully satisfying the demanding power budget for the NUC and manipulator. Mechanically, the introduction of the V2 3D-printed modular payload sled enhanced manufacturability, minimized material usage, and ensured structural integrity, which was further validated through detailed stress and deflection calculations. Furthermore, the software subsystem was upgraded with a robust finite state machine and a highly efficient YOLO11n object detection pipeline, ensuring real-time perception, spatial reasoning, and reliable autonomous navigation.

In strict alignment with the FDR coursework guidelines, this report introduced dedicated analyses for Sustainability and Cyber Security. The comprehensive Sustainability Checklist guarantees that the system adheres to environmental standards across its lifecycle—from eco-friendly material selection and energy-efficient software configurations to waste reduction and repurposing strategies. Simultaneously, the newly integrated Cyber Security framework ensures the integrity of the robot's control logic through physical network isolation, middleware hardening via FastDDS XML, and rigorous multi-level command validation.

Ultimately, as systematically evaluated through the detailed Requirements Verification Matrix (RVM), the integrated Leo Rover platform and myCobot manipulator successfully satisfy all operational, navigational, manipulation, and safety criteria. The final system architecture is structurally sound, computationally efficient, highly secure, and fully prepared for operational deployment and testing.